



Figure 31-8 Dog swimming in a pool.

correlation between a thorough understanding of the dysfunction and good home program compliance.

Treatment

Techniques for performing rehabilitation in the water include interval swimming, free swimming (Figure 31-8), resisted swimming, swimming against a current, swimming that is enhanced by manual techniques, swimming while wearing an adaptive device, or passive techniques performed by the therapist. Interval swimming is performed early in rehabilitation and conditioning and consists of short periods of swimming alternating with short rest periods. This generally results in more rapid progress in strength and cardiovascular conditioning rather than allowing the dog to swim for a longer period until exhaustion. Free swimming often involves following the owner or therapist around the pool or retrieving. When doing resisted swimming the dog is restrained by holding the vest, or using a leash or harness. In enhanced swimming, the therapist provides manual contacts to the feet or extremity to facilitate a stronger contraction or larger range of motion. This technique is reminiscent of proprioceptive neuromuscular facilitation (PNF) used in human physical therapy.⁶

Adaptive devices to enhance swimming include flotation cuffs, small weight cuffs, and elastic bands. These can be attached to different segments of the limb to encourage exaggerated movement or provide added resistance. Facilitation of joint and limb movement particularly in attempting to improve active range of motion is relatively easy in the water. Passive range of motion can easily be performed with the dog either floating in the vest or standing in shoulder deep water. Watsu is a technique in which the therapist supports and moves a relaxed patient along the surface of the water using the resistance of the water to perform passive range of motion on the spine and/or extremities of the patient.³

During the treatment session the therapist must constantly monitor the patient. Be aware of the patient's level of anxiety. Know when to allow the dog to rest. Pushing

the chin down, flattening the ears, and searching the pool perimeter are indicators of stress or fatigue. When exercising the patient look for other signs of fatigue such as a change in the swim stroke or in their breathing. It is also helpful to monitor the patient's tongue and gum color; cyanosis is indicated by a blue color to the tongue and mucous membranes. This is an indication to discontinue exercise immediately and allow a period of rest. Signs of distress previously listed precede this point, and patients that are older, poorly conditioned, or obese should especially not reach this point. It is important to keep patients motivated by changing the routine or positioning them so that they work at a level that maximizes recovery.

Precautions/Contraindications

There are some contraindications for pool rehabilitation. At times it is difficult to rule out if pool therapy is appropriate for a patient without attempting a session to monitor the patient's response. Typical conditions that do not respond well to pool rehabilitation are acute biceps tendonitis and other severe muscle strains as well as early stages of TPLO or fracture repair recovery. Swim therapy should be used with caution or avoided with cardiac conditions that limit exertion levels, especially in warm water. Patients with borderline respiratory conditions may have difficulty because of the hydrostatic pressure placed on the thorax. Patients with open wounds or surface infections should not be placed in pool. Finally, incontinent patients or patients with diarrhea should not undergo swim therapy.

The following case demonstrates the usefulness of aquatic therapy in restoring physical condition to a dog that had lost a great deal of function.

Case Report

Blackie, a 2-year-old hound dog, was presented with a nonunion fracture of the left femur. The fracture was a short oblique fracture of the midshaft of the femur. The original repair consisted of an intramedullary pin and cerclage wire. The fixation devices were prematurely removed 2 months after surgery. Blackie refractured the same bone 4 weeks later. At this time, an external skeletal fixator was placed. Complications associated with pin loosening and continued instability of the fracture site resulted in placement of an intramedullary pin. At this point the stifle was stiff. A recheck examination 2 weeks after surgery indicated that the dog was not using the leg. The dog was barely weight bearing while standing and was non-weight bearing at a walk. Twelve weeks after the last surgery, all fixation devices were removed. Following pin removal instability of the fracture site was palpated.

Seven months after the original fracture Blackie was examined at a surgery referral clinic. At that time the dog would occasionally toe-touch the limb during walking,

but would hold the affected limb up most of the time. In addition the leg was painful, and the stifle had only 5 to 10 degrees ROM. The quadriceps muscles were adhered to the fracture callus and the femur. Adhesions and excessive callus were surgically removed to increase the stifle ROM. A bone plate was applied to stabilize the fracture and align the bone. Following surgery maximum flexion of the knee was 85 degrees and maximum extension was 160 degrees.

The day after surgery a nonsteroidal antiinflammatory drug was administered twice daily. ROM and stretching exercises were performed on the hip, stifle, and hock for 20 repetitions, holding each stretch for 10 seconds. These were performed four to five times daily. Following ROM exercises an ice pack was applied to the area for 10 minutes. Two days after surgery stifle extension was 155 degrees and flexion was 70 degrees. Three days after surgery Blackie was discharged from the hospital with instructions to limit activity to leash walks only and to perform ROM exercises four to five times daily. The therapy sessions were to be preceded by hot packs, and cold packs were to be applied after therapy. Following suture removal, swimming was recommended. The owner was warned that Blackie might never regain full function of the limb because of the stiff stifle.

For 2 months the owner continued performing the ROM exercises and walking. In addition, following suture removal the owner would allow the dog to swim in a river daily. The length of time was gradually increased so that he was swimming for 30 minutes daily by 1 month after surgery.

One month after surgery Blackie returned for a recheck examination. It was noted that the stifle had improved ROM. The thigh muscles were increasing in size, and the dog was using the leg quite well, although there was still an abnormal gait at faster speeds. Radiographically, there was continued but incomplete healing of the fracture site. Blackie was discharged with instructions to continue physical rehabilitation and increase the length of leash walks.

The swimming activity was continued, and the dog was exercised up and down hills by the owner. This activity was continued daily with the amount of hill activity increasing to 1 mile by 8 weeks after surgery.

Twelve weeks after surgery Blackie returned for a recheck examination. At this time Blackie had full flexion (35 degrees) and extension (170 degrees) of the stifle. There was a slight limp at a faster gait, but there were no signs of pain and there was increased musculature of the left rear limb. The fracture had reached union, and return to full activity was recommended.

Although several exercises were used in the management of this case, the early swimming activity was likely instrumental in the recovery of the dog. Because swimming was allowed very soon after surgery the dog was able to actively flex and extend the stifle joint and contract

and relax muscle groups with minimal weight-bearing activity. The gradual reintroduction of weight-bearing activity as the fracture site became more stable enhanced rehabilitation. In this case the rehabilitation was primarily performed by the owner after careful instruction, and with minimal facilities.

REFERENCES

1. Haralson KM: Therapeutic pool programs, *J Clin Manage* 5:510-513, 1988.
2. Hecox B, Mehreteab TA, Weisberg J: *Physical agents: a comprehensive text for physical therapists*, Norwalk, CT, 1994, Appleton & Lange.
3. Bates A, Hanson N: *Aquatic exercise therapy*, Philadelphia, 1996, Saunders.
4. Edlich RF et al: Bioengineering principles of hydrotherapy, *J Burn Care Rehab* 8:579-584, 1987.
5. Skinner AT, Thompson AM, eds: *Duffield's exercise in water*, ed 3, London, 1983, Baillière Tindall.
6. Jamison L, Ogden D: *Aquatic therapy using PNF patterns*, Tucson, AZ, 1994, Therapy Skill Builders.
7. Geigle PR et al: Aquatic physical therapy for balance: the interaction of somatosensory and hydrodynamic principles, *J Aquat Phys Ther* 5:4-10, 1997.
8. Hay J: *The biomechanics of sports techniques*, Englewood Cliffs, NJ, 1978, Prentice-Hall.
9. Roberson JA, Crowe CT: *Engineering fluid mechanics*, Boston, 1985, Houghton Mifflin.
10. McLean SP, Hinrichs RN: Sex differences in the center of buoyancy location of competitive swimmers, *J Sports Sci* 16:373-383, 1998.
11. Harrison RA, Bulstrode S: Percentage weight-bearing during partial immersion in hydrotherapy pool, *Physiother Pract* 3:60-63, 1987.
12. Thein JM, Brody LT: Aquatic-based rehabilitation and training for the elite athlete, *J Orthop Sports Phys Ther* 27:32-41, 1998.
13. Levine D, Marcellin DJ, Millis DL et al: Effects of partial immersion on vertical ground reaction forces and weight distribution in dogs, *Am J Vet Res* 71:1413-1416, 2010.
14. Johns KM: Aquatic therapy: therapeutic treatment for today's patient, *Phys Ther Prod* 24-25, 1993.
15. Bueche F: *Principles of physics*, ed 4, New York, 1982, McGraw-Hill.
16. Babb RW, Simelson-Warr A: Manual techniques of the lower extremities in aquatic physical therapy, *J Aquat Phys Ther* 14:7-15, 1996.
17. Bravo G et al: A weight-bearing, water-based exercise program for osteopenic women: its impact on bone, functional fitness, and well-being, *Arch Phys Med Rehab* 78:1375-1379, 1997.
18. Speer KP et al: A role for hydrotherapy in shoulder rehabilitation, *Am J Sports Med* 21:850-853, 1993.
19. Whitley JD, Schoene LL: Comparison of heart rate responses: water walking versus treadmill walking, *Phys Ther* 67:1501-1504, 1987.
20. Johnson BL et al: Comparison of oxygen uptake and heart rate during exercises on land and in water, *Phys Ther* 57:273-278, 1977.

21. Evans BW, Cureton KJ, Purvis JW: Metabolic and circulatory responses to walking and jogging in water, *Res Q* 49: 442-449, 1978.
22. Hall J et al: Cardiorespiratory responses to underwater treadmill walking in healthy females, *Eur J Appl Physiol* 77:278-284, 1998.
23. Melton-Rogers S et al: Cardiorespiratory responses of patients with rheumatoid arthritis during bicycle riding and running in water, *Phys Ther* 76:1058-1065, 1996.
24. Tovin BJ et al: Comparison of the effects of exercise in water and on land on the rehabilitation of patients with intra-articular anterior cruciate ligament reconstructions, *Phys Ther* 74:710-719, 1994.
25. Kelsey DD, Tyson E: A new method of training for the lower extremity using unloading, *J Orthop Sports Phys Ther* 19:218-223, 1994.
26. Templeton MS, Booth DL, O'Kelly WD: Effects of aquatic therapy on joint flexibility and functional ability in subjects with rheumatic disease, *J Orthop Sports Phys Ther* 23:376-381, 1996.
27. Suomi R, Lindauer S: Effectiveness of arthritis foundation aquatic program on strength and range of motion in women with arthritis, *J Aging Phys Activity* 5:341-351, 1997.
28. Avellini BA, Shapiro Y, Pandolf KB: Cardio-respiratory physical training in water and on land, *Eur J Appl Physiol* 50:255-263, 1983.
29. Perk J, Perk L, Boden C: Cardiorespiratory adaptation of COPD patients to physical training on land and in water, *Eur Resp J* 9:248-252, 1996.
30. Choukroun ML, Varenne P: Adjustments in oxygen transport during head-out immersion in water at different temperatures, *J Appl Physiol* 68:1475-1480, 1990.
31. Turner A, Higgins M, Craddock JC: Disinfection of immersion tanks (Hubbard) in a hospital burn unit, *Arch Environ Health* 28:101-104, 1974.
32. Millis DL et al: A preliminary study of early physical therapy following surgery for cranial cruciate ligament rupture in dogs. Proceedings of the 24th Annual Conference of Veterinary Orthopedic Society, Big Sky, MT, 1997.
33. Taylor RA: Post-surgical physical therapy: the missing link, *Comp Contin Educ Pract Vet* 14:1583-1593, 1982.
34. Millis DL, Levine D: The role of exercise and physical modalities in the treatment of osteoarthritis, *Vet Clin North Am Small Anim Pract* 27:913-930, 1997.
35. Marsolais GS, McLean S, Derrick T et al: Kinematic analysis of the hind limb during swimming and walking in healthy dogs and dogs with surgically corrected cranial cruciate ligament rupture, *J Am Vet Med Assoc* 222:739-743, 2003.
36. Marsolais GS, Dvorak G, Conzemius MG: Effects of post-operative rehabilitation on limb function after cranial cruciate ligament repair in dogs, *J Am Vet Med Assoc* 220: 1325-1330, 2002.
37. Jackson A et al: Joint kinematics of dogs walking on ground and aquatic treadmills. In Proceedings of the 2nd International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine, Knoxville, 2002.



SECTION VI

Physical Therapy for Specific Diagnoses

32 Common Orthopedic Conditions and Their Physical Rehabilitation

Jacqueline R. Davidson and Sharon Kerwin

Shoulder Conditions

Articular Fractures of the Shoulder

It is critical that articular fractures be treated surgically with anatomic reduction and rigid stabilization of the articular surfaces. This is generally achieved using implants such as screws, wires, pins, or small plates. Nonsteroidal antiinflammatory drugs (NSAIDs), other analgesics, passive range of motion (PROM) exercises, and short leash walks are used for the first few weeks after surgery. The goal of the early postoperative period is to maintain and improve joint range of motion (ROM), limit periarticular fibrous tissue, and allow time for adequate bone healing to support more vigorous active weight-bearing exercises. After adequate healing has occurred, physical rehabilitation consists of continued ROM exercises with stretching if needed, and active weight-bearing exercises such as progressive leash walking and aquatic therapy ([Table 32-1](#)).

Avulsion of the supraglenoid tubercle is an injury that occurs in immature dogs and results in displacement of the tubercle. This injury may be classified as a traction physal injury. Affected dogs have an acute non-weight-bearing lameness and pain on palpation of the distal scapula. Surgical repair of this injury involves reduction and stabilization of the supraglenoid tubercle with pins, wires, or screws. Alternatively the fragment may be removed and the biceps tendon left unattached or released. Physical rehabilitation includes NSAIDs, other analgesics, cryotherapy, leash walks, and PROM for the first 3 weeks. After this initial healing period, limb strengthening, especially of the biceps brachii and brachialis muscles, is begun.

Scapular fractures may occur in any breed of dog as a result of trauma. If the neck of the scapula or articular surface of the glenoid is involved, open reduction and internal fixation are usually indicated. Alternatively many nondisplaced or comminuted scapular body fractures are best treated conservatively. In either case it is important to

maintain or regain normal scapular motion in all three planes. Postoperative physical rehabilitation includes NSAIDs, cryotherapy, and PROM exercises to the glenohumeral joint for the first several weeks. As healing progresses scapular PROM is performed in increasing amounts in all planes to prevent fibrosis and decreased ROM. Superficial heat applied with hot packs may be used before exercise to allow increased tissue extensibility, and cross-fiber frictional massage may help to prevent or break down adhesions. Joint mobilization using scapular glides is also effective for increasing ROM. Aquatic therapy (underwater treadmill [UWTM] and swimming), and gait activities such as treadmill walking may also be incorporated to regain muscle strength.

Both conservatively managed and surgically operated articular shoulder fractures are often supported in non-weight-bearing slings postoperatively for several weeks. The small pieces of bone involved combined with a potentially large amount of force normally exerted on the shoulder joint predispose the animal to failure of the repair unless the repair is supported until healing has progressed. Unfortunately, unwanted side effects of non-weight-bearing slings (carpal flexion, Spica, or Velpeau bandages), include loss of nutrition to cartilage, joint stiffness, loss of range of motion, and muscle atrophy. Massage, PROM, and measurement of flexion/extension angles should be performed during each bandage change, and owners warned that several months of rehabilitation may be required after the bandage is removed to re-establish function. Commercially available, removable slings may be more ideal, allowing for some motion without weight bearing and in some cases, removal to allow intermittent, controlled weight-bearing, or rehabilitation exercises on a daily basis.

Osteochondritis Dissecans of the Shoulder

Osteochondrosis is a developmental disease affecting the growth plates and joint cartilage in medium and large breed dogs. Abnormal endochondral ossification of the

Table 32-1 Rehabilitation After Surgical Repair of Articular Fracture

Goals	Treatment
Control inflammation	Cryotherapy immediately after surgery and then daily until inflammation resolves, particularly after exercise
Maintain range of motion and joint health	Passive range of motion and/or joint mobilizations for 3 to 4 weeks. Consider aquatic walking to minimize joint stress. Add cavaletti walking after several weeks, depending on stability of repair.
Control pain	Nonsteroidal antiinflammatories plus other analgesics if needed. ± Transcutaneous electrical nerve stimulation
Strengthen muscle	Begin with controlled slow leash walking. Amount of activity depends on stability of repair and stage of healing. Short, slow leash walks for at least the first few weeks.

deep layers of articular cartilage results in focal areas of thickened cartilage that are prone to injury. Normal activity may cause enough stress to create a fissure in the articular cartilage. Contact of the synovial fluid with the deep layers of degenerating cartilage may cause synovitis. In the absence of further stress, the lesion may heal. However, further stress on the cartilage may result in formation of a cartilage flap. This condition is termed *osteochondritis dissecans* (OCD). Osteochondritis dissecans has been described in various joints of the dog, but occurs most commonly in the caudal aspect of the humeral head. The etiology is multifactorial in all species. Risk factors in dogs include young age, large and giant breeds, rapid growth (high calorie diet), and excess calcium consumption.¹

Dogs are typically presented with mild to moderate weight-bearing lameness between 4 and 8 months of age. Many dogs have bilateral lesions, although one side may be clinically more severe. Muscle atrophy may be apparent in the affected limb if the dog has had a chronic lameness. Pain may be elicited on extreme flexion or extension of the shoulder. Radiographs may demonstrate a defect in the subchondral bone under the cartilage flap on the caudal aspect of the humeral head (Figure 32-1). Uncommonly, advanced imaging or arthroscopy is needed to confirm the diagnosis.²



Figure 32-1 Osteochondritis dissecans of the caudal aspect of the head of the humerus. The right caudal humeral head is flattened and the linear density adjacent to the caudal humeral head is consistent with an osteochondral flap.

If shoulder osteochondrosis is diagnosed early, decreased activity may prevent cartilage flap formation. Cryotherapy and NSAIDs may be used to limit inflammation. Use of disease-modifying osteoarthritis agents (DMOAs), such as glucosamine and chondroitin sulfate, may support cartilage health. Passive range of motion exercises help to maintain joint range of motion and provide joint nutrition and stress along normal lines of movement.³ Once a flap has formed it will not heal and should be removed. If the dog is more than 7 months of age or the lameness has persisted more than 4 to 6 weeks, surgery is indicated.⁴

Surgery is generally the treatment of choice in dogs with clinical signs. It is performed via arthrotomy or arthroscopy to remove the defective cartilage and forage or curettage the bed of the lesion (Figure 32-2). This encourages vascular ingrowth and healing by the formation of fibrocartilage. After surgery NSAIDs, cryotherapy, PROM, and controlled leash walks are instituted for the first 2 to 4 weeks (Table 32-2). After this initial healing period the duration of the leash walks is progressively intensified. In addition treadmill walking, UWTM exercise, and swimming may be initiated. By 6 weeks, light jogging usually can be started.³ Seroma formation is a common complication after shoulder surgery by arthrotomy, but is usually self-limiting and resolves with rest. If a shoulder seroma is observed, the activity level should be reduced until it resolves. The value of treating a seroma with hot packs or cold packs has not been documented, but is commonly recommended. Long-term management of OCD is focused on limiting or treating osteoarthritis (OA). The long-term prognosis for shoulder OCD is excellent in most cases.^{2,5}

Bicipital Tenosynovitis

Bicipital tenosynovitis is a degenerative or failed healing response of the tendon and tendon sheath of the biceps brachii muscle.^{6,7} Inflammation is not consistently present. Tendonitis implies the presence of inflammation in the tendon and musculotendinous junction, and may not be accurate for many cases of tendon pain.^{8,9} Tendinopathy is an inclusive term used to describe tendon pathologies such as tendonitis, tendinosis, or paratenonitis. Tendon injuries may be caused by acute trauma or repetitive loading. Repetitive tendon overload tends to respond with tendon sheath inflammation, tendon degeneration, or both. There is some evidence that acute tendinitis is quickly replaced by tendon degeneration.¹⁰ Tendon pain should not be assumed to be caused by inflammation, but can occur with tendinosis as well.

Bicipital tenosynovitis is commonly seen in middle- to older-age large-breed dogs. Affected dogs usually have

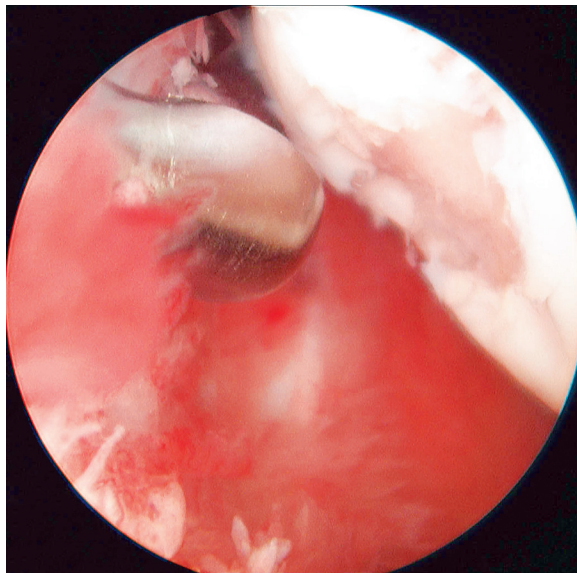


Figure 32-2 Arthroscopic view of osteochondral defect on the caudal humeral head. Subchondral bone is visible in the center of the lesion on the right of the image. The metallic object is a bone curette.

a weight-bearing forelimb lameness that is insidious in onset, progressive or intermittent, and exacerbated by exercise. Pain is not a consistent finding, but may be elicited on direct palpation of the bicipital groove on the proximal craniomedial aspect of the humerus. The pain response may be increased by flexing the shoulder with simultaneous elbow extension, which increases biceps tendon tension. Bicipital tenosynovitis may occur in conjunction with OA of the shoulder or glenohumeral ligament damage. Atrophy of the supraspinatus and infraspinatus muscles may be palpable, particularly in chronic cases. Mineralization of the biceps tendon, osteophytes in the intertubercular groove, or other degenerative changes may be seen on radiographs. Diagnosis can be difficult and is sometimes made by diagnostic ultrasound, magnetic resonance imaging (MRI), or observation during arthroscopy.

Conservative management of tendinopathies or tendinitis in humans primarily consists of rest, icing, and eccentric strengthening exercises. There are little data to support the use of therapeutic ultrasound, shock wave therapy, orthotics or massage,⁸ but these are used in some cases. Analgesics are indicated, but not necessarily NSAIDs or corticosteroids. Deep transverse friction massage has been advocated as part of the treatment for tendinopathy pain. Studies of the effectiveness of deep transverse friction massage for the treatment of tendinopathies in humans are inconclusive.¹¹ Common recommendations for the conservative management of tendinopathies in dogs include 4 to 12 weeks of rest and NSAIDs as needed (Table 32-3). As an alternative to oral NSAID therapy, intraarticular corticosteroids may be administered because the shoulder joint communicates with the biceps tendon sheath. Pulsed-mode 3-MHz therapeutic ultrasound is advocated over the tendon and musculotendinous junction to promote tissue healing. Because of the close proximity to bone, the dog must be monitored for any signs of pain, and the ultrasound intensity adjusted accordingly. Pulsed-mode ultrasound should not result in significant tissue temperature increase. Laser therapy may also be used to decrease pain and aid healing. Cryotherapy may be prescribed to reduce inflammation and pain. Response to conservative therapy varies, and may

Table 32-2 Joint Rehabilitation After Osteochondritis Dissecans Surgery

Goals	Treatment
Control inflammation	Cryotherapy immediately after surgery and then daily until inflammation resolves, particularly after exercise
Maintain range of motion and joint health	Passive range of motion for 1 to 2 weeks
Control pain	Nonsteroidal antiinflammatories for first few weeks, as needed
Strengthen muscles	Begin with controlled slow leash walking and progress over 2 to 4 weeks. Decrease activity if seroma or lameness develops. May begin jogging or swimming in 6 weeks.

Table 32-3 Rehabilitation for Tendinopathies

Goals	Treatment
Control inflammation	Cryotherapy immediately after surgery and then daily until inflammation resolves, particularly after exercise
Promote tissue healing	Limited activity $\pm$ pulsed ultrasound $\pm$ cross-fiber frictional massage
Control pain	Analgesics $\pm$ transcutaneous electrical nerve stimulation $\pm$ laser
Strengthen soft tissues	May need several weeks of rest after acute injuries or surgery. Gradual progression of strength exercises, starting with eccentric exercises for involved muscles, such as walking downhill.
Maintain joint range of motion	Joint mobilizations or passive range of motion

be related to the inciting cause of the tendinopathy.^{12,13} Full recovery can be expected to require 3 to 6 months of therapy.⁸

Surgical treatment is recommended for dogs that do not respond to medical treatment or have significant tears of the tendon. Surgical techniques include tenodesis or tendon release, both of which have been described as arthroscopic or open procedures.^{12,14-16} Cryotherapy, PROM, and short leash walks (preferably with a harness that does not rub over the surgical site) are used for the first 3 postoperative weeks to promote normal joint movement while the tendon has minimal strength. Joint mobilizations may also help increase joint ROM. Exercises to gradually increase limb strength and promote remodeling of the peritendinous scar are performed after 3 weeks. Exercises should emphasize strengthening of the brachialis muscle, to improve the dog's ability to flex the elbow joint. Shake hands exercises with limb weights that are gradually increased, as well as walking over cavaletti rails with limb weights, are specific exercises to increase the strength of elbow flexor muscles. Aquatic therapy and treadmill exercises may be beneficial. Superficial heat therapy may be used before exercise to facilitate tissue extensibility. Neuromuscular electrical stimulation may be applied to the brachialis muscle. Cross-fiber frictional massage can be used to prevent or diminish adhesions. Cryotherapy may help to minimize postexercise inflammation. Most surgically treated dogs regain near normal function and gait.^{12,13}

Supraspinatus Tendinopathy

Supraspinatus tendinopathy may cause mild to moderate lameness in medium and large breed dogs. The etiology of supraspinatus tendinopathy is unknown, but it is speculated to be caused by overuse and repetitive trauma.¹⁷ Affected dogs typically have chronic, intermittent, unilateral lameness. Palpation of the greater tubercle, near the tendon insertion, is usually painful. Mineralization may be observed on radiographs, and may contribute to bicipital tenosynovitis or impingement of the biceps tendon. However, the presence of mineralization can also be asymptomatic and may be an incidental finding. Diagnostic ultrasound or MRI may be needed to identify non-mineralized tendinopathy. Medical treatment includes rest, PROM, NSAIDs, local steroid injection, heat or cold therapy, deep massage therapy, and therapeutic ultrasound. Response to these treatments is limited. However, pulsed therapeutic ultrasound resulted in some clinical improvement in one study of calcific tendonitis in humans.¹⁸ Extracorporeal shock wave therapy may decrease the size of the mineralized region in dogs.¹⁹ If medical treatment is inadequate, surgical excision of the mineralized tissue is recommended.^{17,20-22} After surgery, a carpal flexion bandage may be applied for 2 weeks to prevent weight bearing. Activity is limited for an additional 2 to 3 weeks. Swimming may cause too much tendon stress and is not advised for several months. The prognosis varies with the severity of the mineralization.¹⁷ Most dogs will be sound within 3 to 4 weeks. Lameness may recur with persistent bicipital tenosynovitis or recurrence of supraspinatus mineralization.

Medial Shoulder Instability

Medial glenohumeral instability is thought to result from chronic repetitive activity or overuse, such as jumping. Dogs typically develop a chronic, intermittent weight-bearing lameness that is nonresponsive to treatment with NSAIDs. Instability may be assessed subjectively by palpating for increased translocation of the humerus.²³ Medial instability can be confirmed by goniometric measurements of shoulder abduction,²⁴ although the clinician should consider the role of pain-related muscle atrophy as a potential cause for increased shoulder abduction as opposed to true instability of the shoulder. The normal abduction angle in a large breed dog is approximately 30 degrees.²⁴ The affected shoulder will have a much larger abduction angle as compared with the normal shoulder.²⁴ The degree of instability may be characterized as mild, moderate, or severe.²⁵ Mild instability is associated with minimal pathologic changes and abduction angles of 35 to 45 degrees. Dogs with moderate instability have abduction angles of 45 to 65 degrees and more advanced pathology is seen arthroscopically. Dogs with severe instability typically have shoulder abduction angles greater than 65

degrees. Severe disruption of the medial glenohumeral ligament, subscapularis tendon, and joint capsule are seen arthroscopically.²⁵

Conservative treatment with the shoulder supported and restricted by shoulder hobbles can be used in mild cases. Extracorporeal shockwave treatment to the medial shoulder structures may be beneficial. As healing progresses, strengthening of the medial supporting muscles may be achieved by walking on a treadmill with lateral tension supplied by an elastic band during the swing phase of gait. As strength improves progressively stiffer elastic bands may be used. Dogs with moderate shoulder instability are treated arthroscopically with imbrication, thermal capsulorrhaphy, reconstruction of the medial glenohumeral ligament, or transposition of the biceps tendon.^{26,27} After surgery a shoulder support wrap with hobbles is applied to protect the surgical repair. Cryotherapy, PROM, and strictly controlled activity are instituted for the first 3 post-operative weeks.²⁵ After 3 weeks weight-bearing exercises are gradually introduced. Function improves within three to four months, and optimal function is not attained until 5 to 6 months following thermal capsulorrhaphy.²⁶ Severe shoulder instability is usually caused by trauma and is treated as for acute shoulder luxation.

Shoulder Luxation

Shoulder luxation is usually a result of trauma, but may also be seen with congenital joint malformation. Shoulder luxation may be apparent on physical examination, but radiographs are recommended to evaluate for concomitant fractures or congenital bony malformations. If it is a recent injury with no evidence of fracture or malformations, closed reduction and immobilization in a sling for 1 to 3 weeks may be successful. Medial shoulder luxations are immobilized with the limb flexed in a Velpeau sling, whereas lateral luxations are immobilized with a spica splint. After the sling is removed, gentle PROM is initiated. Motion is limited to the sagittal plane to avoid excess stress on the affected joint capsule and ligament. Weight-bearing exercises are also limited.

If conservative management is unsuccessful or fractures are present, surgical stabilization is indicated. Surgical techniques for medial shoulder luxation include imbrication or reconstruction of the medial glenohumeral ligament and joint capsule, or transposition of the biceps tendon to a more medial position to assist with joint stability. Post-operatively treatment with cryotherapy and NSAIDs can be initiated to minimize inflammation, and the shoulder may be placed in a Velpeau sling for 2 weeks. After the Velpeau sling is removed the front limbs may be placed in hobbles to prevent abduction of the limb, but ROM must be restored. Passive ROM exercises are performed, but are limited to the sagittal plane (flexion and extension) to avoid stressing the medial or lateral joint capsule. Exercise is initially restricted, with gradual progression of

weight-bearing exercises. Full recovery may require 1 to 3 months.

The prognosis is generally good, although OA may develop. In the case of severe congenital malformations or irreparable joint fractures, a salvage procedure may be indicated. Salvage procedures of the shoulder include excision arthroplasty (glenoid excision), arthrodesis, or amputation.

Fibrotic Contracture of the Infraspinatus Muscle

Contracture is a shortening of the tendon-muscle unit that is not caused by active muscle contraction. Most or all of the muscle and/or associated tendon is replaced by fibrous tissue. Most cases of contracture have a history of acute injury with lameness and inflammation that subsides with supportive care. Weeks to months later the animal is presented with clinical signs, which may include pain, weakness, firmness within the muscle, and an abnormal posture and gait.²⁸ Proposed causes include infection, trauma, repetitive strains, fractures, infectious diseases, immune-mediated diseases, ischemia, and eosinophilic myositis. Muscle contracture leads to permanent shortening of the affected muscle and can limit ROM of the associated joint. Muscle contracture can occur in any muscle, but has been reported in the infraspinatus, supraspinatus, teres minor, brachialis, quadriceps, semitendinosus, gracilis, and sartorius.²⁸ If the contracture is not severe, splinting with the muscle and tendon in tension and stretching activities may lengthen the tendon-muscle unit. In many cases, surgery is recommended to lengthen or transect the tendon.

Contracture in the front limb has been most commonly reported in the infraspinatus muscle. Infraspinatus contracture causes a mild weight-bearing lameness mainly in hunting or working dogs. The cause is hypothesized to be acute muscle trauma, which results in incomplete rupture of the infraspinatus muscle. The injury leads to inflammation and subsequent compartment syndrome.²⁸⁻³⁰ Replacement of muscle fibers by fibrous tissue occurs over days to weeks. The supraspinatus and deltoid muscles also may be involved and undergo atrophy or contracture.³¹ The humerus is held in adduction, whereas the antebrachium is externally rotated with the foot abducted. The scapulohumeral joint cannot fully extend. There is a weight-bearing lameness in which the limb circumducts as it is advanced during the stride and the paw flips into extension before foot placement.

Physical rehabilitation may be beneficial if the condition is diagnosed early. Continuous mode therapeutic ultrasound with stretching exercises may help to lengthen contracted tissues, but the degree of contracture is usually so severe at the time of diagnosis that it is difficult to improve the condition.³ When the dog is presented with contracture, the treatment of choice is surgical transection of the infraspinatus tendon and associated fibrous tissue.³¹

This immediately restores motion of the shoulder joint and the prognosis is good.

Postoperatively full weight bearing is allowed, but activity should be restricted for the first few weeks. Uncontrolled activity may cause tissue damage and recurrence of fibrous tissue.³ Passive ROM exercises to the forelimb joints several times daily maintain joint ROM and promote normal alignment of the healing tissues.³ Cryotherapy may be used to reduce inflammation after exercise. In cases in which there is significant disuse atrophy of the forelimb muscles, general conditioning exercises for the limb are used to gradually return the muscle to normal size and strength. Conditioning exercises include walking, jogging, and aquatic therapy. Neuromuscular electrical stimulation (NMES) may be used if the atrophy is severe and the dog is too weak to use the leg well.

Teres Minor Strain

Teres minor strain has been reported in athletic dogs of various breeds.³² All dogs in one report had front limb lameness that was improved by rest, but recurred with return to activity. The shoulder was painful on full extension and pain was elicited with direct palpation of the teres minor muscle. Treatment included ultrasound, stretching, balancing exercises, leash walking, gait retraining, and acupuncture point stimulation. All dogs returned to previous levels of activity.³²

Elbow Conditions

Articular Fractures of the Elbow

Fractures of the articular surface of the distal humerus are common and typically result from trauma. Articular fractures of the proximal radius and ulna are much less common. Salter-Harris type IV fractures of the lateral condyle of the humerus are commonly seen in skeletally immature dogs, and fractures of the lateral humeral condyle are also seen in older dogs. In spaniel breeds, such as Cocker spaniels, it is not uncommon for mature animals to present with articular fractures with minimal evidence or history of trauma (Figure 32-3). These dogs appear to have incomplete ossification of the humeral condyles, although there may be other predisposing factors.³³ As with any articular fracture, decreased ROM and destruction of articular cartilage may be sequelae of an articular fracture. In addition Salter-Harris fractures in immature dogs may have alteration in bone growth that is necessary for a congruent elbow joint. Repair is accomplished with placement of a transcondylar screw with compression of the fracture site and placement of a small antirotational pin driven up the lateral epicondyle (Figure 32-4).

Articular fractures demand both rigid fixation and anatomic reduction to maintain an even, stable cartilage surface. Because of the necessity for anatomic reduction,

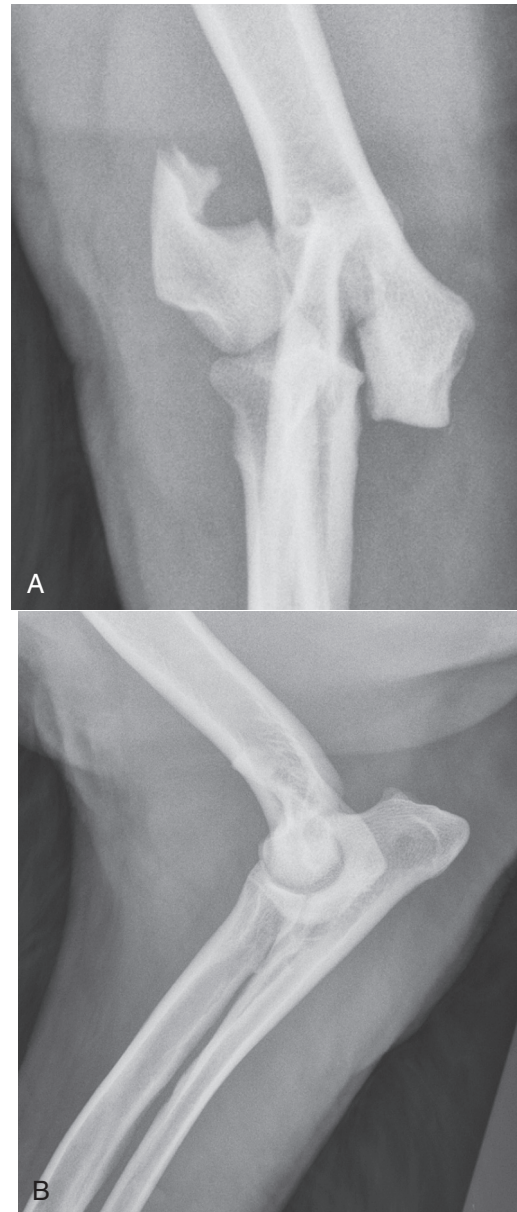


Figure 32-3 Fracture of the lateral condyle of the humerus in a cocker spaniel. **A**, A-P view, **B**, Lateral view.

the surgical approach may be extensive, including tenotomies or osteotomies. The surgeon must carefully balance the degree of soft tissue manipulation with the need for exposure of the fracture to facilitate reduction. Extensive dissection increases postoperative pain and swelling, and may increase the degree of muscle scarring and periarticular fibrosis after surgery. In our experience excessive postoperative scarring and fibrosis leading to decreased range of motion are particularly likely after articular fractures involving the distal humerus or distal femur in the dog and cat. The use of minimally invasive techniques, such as closed reduction and fixation via the use of fluoroscopy, or

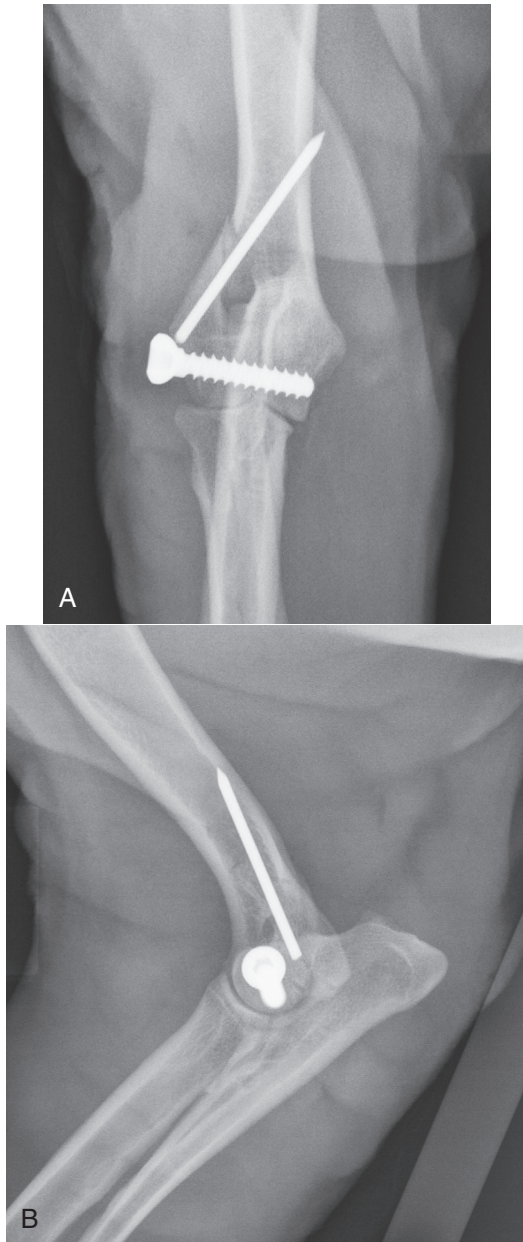


Figure 32-4 Repair of the fracture shown in [Figure 32-3](#) with a transcondylar screw placed in lag fashion to compress the fracture fragments and a small pin in the lateral epicondyle to counteract rotational forces. **A**, A-P view, **B**, Lateral view.

arthroscopic-assisted fracture reduction should be considered ([Figure 32-5](#)).³⁴ Whenever possible, osteotomies rather than tenotomies should be performed to gain access to the fracture because osteotomies heal by reforming normal bone, whereas tenotomies heal via scar tissue. When osteotomies are repaired the surgeon must pay particular attention to pin and wire placement. Inadvertent wire placement into the joint causes mechanical loss of range of motion and pain for the patient, rendering even the most sophisticated rehabilitation strategy useless.



Figure 32-5 Lateral postoperative radiograph of a radial fracture that was plated using a limited approach to minimize disruption of the soft tissues.

In addition to the demand for anatomic reduction, rigid fixation of articular fractures may rely on implants placed across a small epiphyseal segment, resulting in a somewhat tenuous repair, especially in young patients with soft bone. In the case of distal humeral fractures, particularly Salter type IV fractures in young dogs, the surgeon may be tempted to splint the limb or place a carpal flexion bandage to prevent weight bearing. Although these strategies may effectively prevent mechanical overload of the repair during the healing process, they may have the unintended effect of promoting muscle scarring around the operative site as well as cartilage atrophy.³⁵ Although a carpal flexion bandage still allows some ROM exercises, contracture of muscles around the carpus may lead to longstanding pain, loss of ROM and poor limb use despite good fracture healing. ROM exercises should be performed frequently but are best left to professionals as excessive (or too little) ROM can cause failure of fixation or joint contractures respectively. Weight-bearing exercises should be used with extreme caution, especially during the first 3 weeks after repair to avoid fixation failure. Short leash walks may be permitted, but it is important to properly confine the patient to avoid catastrophic failure, especially if the patient appears to be doing very well.



Figure 32-6 Therapy ball may be used to encourage elbow flexion.

Cryotherapy, analgesic medication, massage, and passive ROM may be used for articular fractures of the elbow (see [Table 32-1](#)). In small breed dogs use of a therapy ball may be helpful to encourage limb use and flexion of the elbow without direct manipulation of the limb ([Figure 32-6](#)). It is imperative that ROM exercises continue for 3 to 4 weeks after surgery to be certain that ROM is maintained. Normal ROM may return 10 to 14 days after surgery, but fibrous tissue and other healing periarticular tissues are not yet mature. If ROM exercises are discontinued too soon, it is possible that some maturation and contracture of tissues may occur with some loss of normal joint motion. These fractures can present a very difficult challenge to the surgeon and the rehabilitation practitioner. Although overuse may result in catastrophic failure of the repair, failure to perform appropriate ROM exercises may lead to crippling loss of ROM.

An even more challenging fracture to repair and rehabilitate is a bicondylar (sometimes referred to as a T or Y) fracture, in which both the medial and lateral condyles of the humerus are fractured from the main shaft of the humerus and from each other. It is fairly common for several additional fragments to be separated from the shaft of the humerus. Repair involves stabilization and fixation of the condyles, and then stabilizing them to the main shaft of the humerus with plates, intramedullary pins, or an external fixator. In many instances, an osteotomy of the olecranon is necessary to repair this type of fracture. The olecranon osteotomy is then repaired with a pin and tension band or screw technique. Decreased ROM is very common, especially after an olecranon osteotomy, along with OA development after surgery. Newer techniques that use separate medial and lateral approaches to the joint to avoid an olecranon osteotomy seem to result in maintenance of

greater ROM. It is critical that rehabilitation be performed following repair of bicondylar fractures of the humerus. Particular attention is paid to achieving as complete ROM as possible, using passive and active ROM exercises. Control of periarticular swelling and edema with cryotherapy, NSAIDs, and bandages helps to improve motion and reduce postoperative pain. Swimming or walking in an underwater treadmill has the additional benefit of limb usage in a buoyant environment. As weight-bearing improves, other activities, such as walking over cavaletti rails may be initiated. Daily rehabilitation for at least 2 to 3 weeks is recommended.

Avulsion fractures of the olecranon may also occur. In immature dogs, there may be an avulsion fracture of the olecranon with separation across the physis. Olecranon fractures prevent extension of the elbow, which is necessary for normal weight bearing and ambulation. This condition is treated by surgically replacing and stabilizing the avulsed fragment with a screw or pin and tension band technique, or if the fragment is large enough, a bone plate. Fracture through the articular surface of the ulna may also occur, necessitating excellent anatomic reduction and rigid internal fixation, generally with a bone plate and screws. Physical rehabilitation is important to achieve optimal elbow extension. Treatments in the first 3 weeks may include gentle PROM, NSAIDs, and cryotherapy. Following adequate healing more vigorous ROM and stretching exercises may be initiated, along with active exercises to rebuild atrophied muscles.

Elbow Dysplasia

Elbow dysplasia includes fragmented medial coronoid process (FCP), osteochondritis dissecans (OCD), and ununited anconeal process (UAP). Dogs with elbow dysplasia usually have only one of the three conditions. Elbow incongruity may also be present. The pathogenesis is poorly understood. Factors that have been implicated include genetics, improper nutrition, and trauma. Other proposed mechanisms are osteochondrosis, trochlear notch dysplasia, and asynchronous growth of the radius and ulna.³⁶ Elbow dysplasia usually occurs in large or giant breeds of dogs. Affected dogs typically have bilateral problems, although one elbow may be more severely affected.

Fragmented Medial Coronoid Process of the Ulna

Dogs with FCP are presented with a mild to moderate weight-bearing lameness, which is usually noted between 4 and 7 months of age.³⁷ Physical exam findings include pain on flexion and extension of the elbow, pain on palpation of the medial aspect of the joint, and palpable joint effusion. In dogs older than 11 months, crepitus, decreased ROM, and general joint thickening may be evident. Radiographs often show degenerative changes in the joint, but the actual lesion may not be seen ([Figure 32-7](#)). In many

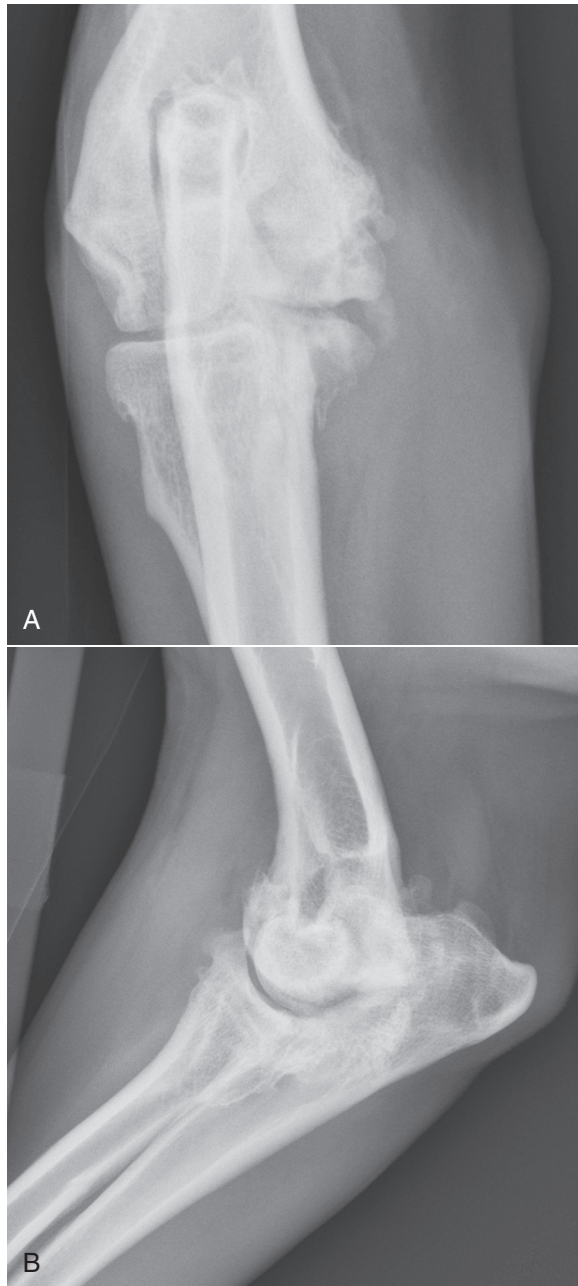


Figure 32-7 Severe elbow osteoarthritis in a middle-aged dog suspected to be secondary to a previously unrecognized fragmented medial coronoid process. **A**, A-P view, **B**, Lateral view.

cases, computed tomography or arthroscopy is needed to make a definitive diagnosis.

Treatment is removal of the abnormal process via arthrotomy or arthroscopy, with possible subtotal coronoidectomy or ulnar osteotomy in some cases. Postoperatively activity is limited for 2 to 4 weeks. The prognosis is good if FCP is removed before there is advanced OA. Osteoarthritis will progress regardless of treatment, but changes are thought to be more severe in untreated cases. If the OA is severe at the time of diagnosis, the value of

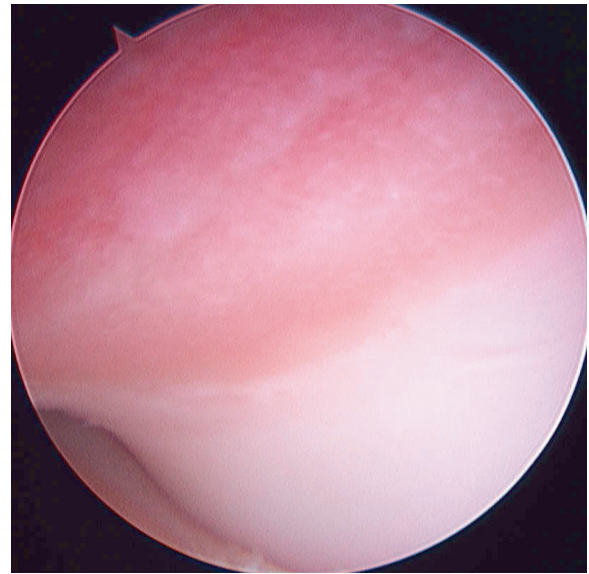


Figure 32-8 Arthroscopic view of an erosion on the humeral articular cartilage that was created by a fragmented medial coronoid process. This is known as a kissing lesion and may be confused with osteochondritis.

surgical treatment must be questioned. These dogs may be better managed by conservative treatment for the OA.³⁷ In all cases rehabilitation is directed toward preventing, slowing, or treating OA to maintain an acceptable quality of life. Therapy includes maintaining an appropriate body weight, controlled low-impact exercise, aquatic exercise, monitoring pain, NSAIDs, and chondroprotective agents. See section on OA for guidelines on rehabilitation.

Osteochondritis Dissecans of the Elbow

Dogs with elbow OCD are typically presented with mild to moderate lameness between 4 and 9 months of age. Many dogs have bilateral lesions, although one side may be clinically more severe. Atrophy of the muscles may be apparent in the affected limb if the dog has been lame for several weeks. Pain may be elicited on extreme flexion or extension of the affected joint. Chronic elbow OCD may also have signs of OA such as joint effusion, thickening of the periarticular soft tissues, decreased ROM, and crepitus. Radiographs may demonstrate a defect in the subchondral bone under the cartilage flap on the medial humeral condyle. Fragmented medial coronoid process can create an erosive lesion on the medial humeral condyle, called a *kissing lesion*, which can look very similar to OCD at the time of surgery (Figure 32-8). In some cases especially with elbow OCD, the diagnosis may be difficult and is best made by computed tomography or arthroscopy.

Surgery is usually the treatment of choice in dogs with clinical signs. It is performed via arthrotomy or arthroscopy to remove the defective cartilage and forage or curettage of the lesion. After surgery NSAIDs, cryotherapy,



Figure 32-9 Ununited anconeal process. An irregular radiolucent line is visible between the olecranon and the anconeal process on a flexed, lateral radiograph of the elbow, indicating that the anconeal process has not fused with the ulna.

PROM, and controlled leash walks are instituted for the first few weeks (see [Table 32-2](#)). After the initial healing period, the duration of the leash walks are progressively intensified. Long-term management of OCD is focused on limiting or treating OA. See section on OA for guidelines on rehabilitation. The prognosis for elbow OCD is good if surgery is performed before OA is advanced, but it is more guarded than for OCD of the shoulder.

Ununited Anconeal Process

Ununited anconeal process (UAP) is a failure of the anconeal process to fuse with the olecranon by 5 months of age, and is apparent on a flexed lateral radiograph ([Figure 32-9](#)).³⁷ Instability of the anconeal process causes inflammation and eventual OA. The dog usually is presented with a weight-bearing lameness. Decreased range of motion and joint effusion of the caudolateral aspect of the elbow may be apparent on palpation.

Treatment options include surgical removal of the anconeal process, screw fixation combined with osteotomy of the proximal ulna, or osteotomy of the proximal ulna alone to relieve joint incongruity and allow healing of the anconeal process. Regardless of the treatment, OA progresses, therefore ongoing treatment is directed at managing OA. (See the section on OA for guidelines on rehabilitation.)

If the UAP is treated by screw fixation, the rehabilitation program should have a slower progression of weight-bearing activities until there is radiographic evidence

of healing. This may take as long as 12 weeks in some cases. If the UAP was treated by removal, the rehabilitation program may progress more rapidly, as dictated by the degree of joint effusion, pain, ROM, and weight bearing. Immediately after surgery PROM, aquatic therapy, and light leash walks are recommended. Cryotherapy and NSAIDs may be used as needed to control inflammation and pain. Aquatic activity can be very beneficial because of the active ROM while weight bearing in a buoyant environment.^{38,39}

Elbow Incongruity

Asynchronous growth of the radius and ulna can cause elbow incongruity in chondrodystrophoid dogs, and is usually evident by 4 to 5 months of age. Ununited anconeal process and FCP have been associated with elbow incongruity, although they can occur in the absence of incongruity. Traumatic closure of a radial or ulnar physis also leads to asynchronous growth with elbow subluxation and angular limb deformity.

Surgical options to improve congruity include corrective osteotomy or ostectomy of the radius or ulna, and may involve stabilization with a bone plate or external fixator. The prognosis is worse if the incongruity is severe and the animal is older (>9 months of age).⁴⁰

After surgery normal ROM of the joints proximal and distal to the repair should be maintained. Pain and edema may be controlled by NSAIDs and cryotherapy. If the goal was to lengthen the limb following surgery, increased tension on the soft tissues may result in decreased ROM, joint stiffness, and lameness. In these cases therapeutic ultrasound with simultaneous stretching and ROM activities may be helpful. As bone healing progresses more aggressive weight-bearing exercises may be initiated.

Elbow incongruity invariably results in some degree of OA of the elbow. Some dogs may not present for rehabilitation until they have end-stage OA. In either case rehabilitation is directed at managing OA so as to maintain an acceptable quality of life. (See the section on OA for guidelines on rehabilitation.) Conservative management of OA includes NSAIDs, chondroprotective agents, cryotherapy, PROM, and physical rehabilitation. Dogs with end-stage OA often have chronic synovial effusion, reduced ROM, crepitus of the joint, pain on manipulation of the elbow, muscle atrophy, and lameness. Physical rehabilitation cannot correct the underlying problem, but efforts to improve joint health, increase ROM, reduce weight, and rebuild atrophied muscles may significantly improve the patient's quality of life. Physical rehabilitation considerations would include PROM and stretching, active ROM activities, redeveloping muscle strength with exercise or neuromuscular electrical stimulation, low-impact exercises, laser, extracorporeal shock wave, and aquatic therapy. Careful monitoring of ROM is important with a home program to avoid permanent losses in ROM.

Elbow Luxation

Elbow joint luxations are generally a result of trauma, although congenital malformations of the elbow can predispose to luxation. Traumatic elbow luxations are usually lateral and can be diagnosed by palpating the radial head laterally displaced relative to the humeral condyle. Radiographs are recommended to assess for concomitant fractures and for congenital malformations. Radiographic findings may affect the treatment choice and the prognosis.

If there are no congenital malformations or concurrent fractures, closed reduction of the elbow is usually successful because the joint is inherently quite stable, despite significant collateral ligament damage. Open reduction and surgical stabilization are indicated if closed reduction is not achieved, closed reduction fails, or congenital abnormalities or articular fractures are present. Surgical repair often involves direct repair of the collateral ligaments or replacement of the collateral ligaments with a suture prosthesis (e.g., heavy suture secured by bone anchors or screws). After closed or open reduction, joint healing is protected by maintaining the elbow in an extended position. This is accomplished by placing the limb in a spica splint for 1 to 4 weeks.

Regardless of whether closed or open reduction is performed, rehabilitation is begun after the spica splint is removed. Range of motion must be restored without causing the joint to reluxate. Cryotherapy and NSAIDs are used to minimize inflammation in the early postoperative period. After elbow luxation, passive and active ROM exercises should be limited to the sagittal plane, and internal, external, varus or valgus stresses should be minimized. Activity is gradually increased, but aggressive activities such as jogging, swimming, or uncontrolled play are limited until the joint has healed. This may take 1 to 3 months, depending on the degree of tissue damage. The prognosis for traumatic joint luxations is generally good, although OA is a common sequela.

Joint reduction and preservation of the joint may not be recommended if there are irreparable fractures involving the joint or severe congenital malformations. In these cases a joint salvage procedure, such as elbow arthrodesis, may be a better choice. Total elbow replacement surgery is relatively new in dogs and is not commonly performed, but may result in better function.⁴¹ Limb amputation may be elected in some cases.

Antebrachial Conditions

Fractures of the Antebrachium

Fractures of the radius and ulna commonly occur. These injuries may be treated by closed reduction and immobilization in a cast in cases of simple fractures, or they may be repaired by open reduction and fixation with wires,



Figure 32-10 A, Distal radial fractures in toy breed dogs that are treated with external coaptation (splint or cast) have an increased risk of nonunion. B, In some cases, the radius and ulna may atrophy.

plates, screws, or external skeletal fixation. In small dogs distal radius and ulna fractures have a high incidence of delayed or nonunion healing when treated with external coaptation (Figure 32-10). For optimal healing these fractures require anatomic reduction and rigid surgical fixation, while using appropriately sized implants to reduce the risk of stress protection. In addition to problems with bone healing, prolonged immobilization by splinting or casting leads to disuse atrophy of bone, cartilage, ligaments, tendons, and muscle in the affected limb. In some cases there is frank loss of bone distally in response to disuse of the leg. Following the first few weeks of physical rehabilitation, which involves reduction of pain and swelling, rehabilitation efforts can be undertaken to increase weight-bearing activities to load the bone during fracture healing and help stimulate normal healing. One of the main impediments to limb use is decreased ROM of the carpus as a result of flexural contracture of the soft tissues. Assisted extension of the carpus to a normal position is often painful and often results in a non-weight-bearing lameness, despite healing of the fracture. If a coaptation device is used, it is essential that particular attention be paid to stretching and ROM exercises of the carpus and digits. In some cases, an extension splint molded to the normal contralateral side may be placed on the affected limb, applying some tension to position the digits and metacarpals more normally in between rehabilitation sessions. It may take 10 to 14 days to achieve a normal angle of the carpus for weight bearing.

Angular Limb Deformities

Angular limb deformities can occur as a developmental abnormality or as a result of trauma to a physis. The most common deformity is carpal valgus as a result of fracture of the distal antebrachium. As the animal continues to grow, it may develop subluxation of the elbow and carpal joints with subsequent OA in these joints. Correction of angular limb deformities involves a corrective osteotomy of the affected bone(s) and stabilization with a bone plate or external fixator. In some cases, a circular fixator is used to gradually correct the angular deformity and lengthen the bones by distraction osteogenesis. After surgery cryotherapy and NSAIDs may be used to limit inflammation and pain. It is important to maintain normal ROM of the joints proximal and distal to the repair. This is particularly important for animals undergoing distraction osteogenesis, because the concurrent stretching of soft tissues in the region may cause reduced joint ROM, stiffness, and lameness. Stretching and ROM activities with simultaneous application of heat or therapeutic ultrasound may be beneficial. As bone healing progresses the intensity or duration of aquatic therapy and weight-bearing therapeutic exercises may be increased.

Contracture of the Flexors of the Antebrachium

After limb immobilization or prolonged disuse, the antebrachial flexor muscles may undergo contracture (shortening of the tendon-muscle unit). The result is limited extension of the joints distal to the elbow, which can impair normal weight-bearing. Methods of lengthening and stretching the contracted tendon-muscle units include 3-MHz therapeutic ultrasound or hot packs, manual stretching, exercises, and massage.⁴²⁻⁴⁴ If stretching is too aggressive it may cause damage to soft tissues or bone fractures. A prolonged stretch may be applied with a custom splint on the distal limb. The splint should be applied intermittently to minimize the risk of pressure injuries from the splint. The splint may be applied for longer periods each day until it can be worn all day, if desired. The goal is to stretch and realign the tissues without damaging them.

Carpal, Metacarpal, and Phalangeal Conditions

Carpal, Metacarpal, and Phalangeal Fractures

Fractures of the carpal bones are rare but may involve the accessory carpal bone or the radial carpal bone in racing athletes such as greyhounds. In other breeds carpal fractures are often the result of automobile or other traumatic insults. Some carpal fractures may result in medial or lateral instability if the fracture involves the origin or insertion of the respective collateral ligament. A relatively common injury is a shearing injury, in which the dog's

limb becomes trapped beneath a skidding automobile tire. The soft tissues including skin, subcutaneous tissues, tendons, ligaments, and in severe cases, bone and tendon, are eroded away, leaving the joint unstable. Many of these injuries cannot be primarily repaired, and an arthrodesis is performed in these cases. Fortunately, function is usually very acceptable after bony fusion has occurred.

Metacarpal and phalangeal fractures usually occur as a result of trauma, and phalangeal fractures are often open fractures (Figure 32-11). Carpal, metacarpal, and phalangeal fractures may be treated using closed reduction and immobilization in a coaptation device, or with a variety of implants including intramedullary pins, wires, and plates and screws (Figure 32-12). A splint or light cast is usually applied to the limb after internal fixation to protect the relatively small implants that must be used. If a cast or splint is used elbow ROM must be preserved during the time of coaptation application, and after coaptation removal. Sagittal plane ROM exercises and joint mobilization work well to restore ROM to the area. In cases of fracture and carpal instability, splinting is often used to protect the repair. If possible removing the splint on a weekly basis to perform ROM exercises to maintain joint and tendon mobility is advised. The splint is reapplied to protect the fracture after performing ROM exercises.

Carpal Hyperextension

Carpal hyperextension injuries occur after jumping or falling from a height, spraining the palmar fibrocartilage and the carpal ligaments. Damage may occur to antebrachiocarpal, middle carpal, carpometacarpal joints, or any combination of these joints. Carpal hyperextension



Figure 32-11 When all four metatarsal or metacarpal bones have fractured, surgical repair is generally preferable to external coaptation. **A**, A-P view, **B**, Lateral view.

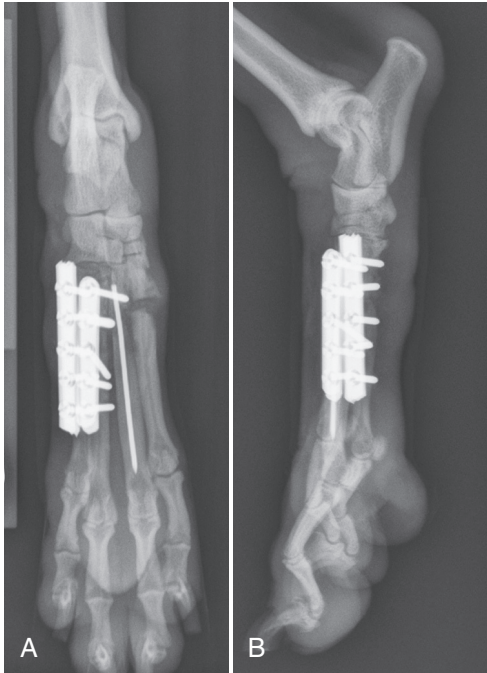


Figure 32-12 Metatarsal fractures that have been stabilized with bone plates and intramedullary pin. A, A-P view, B, Lateral view.

may also be caused by immune-mediated joint disease, or endocrinopathy such as hyperadrenocorticism. Affected animals are lame and walk with the carpus in hyperextension. In severe cases, the carpus may touch the ground. In general healing of a sprain (ligament injury) may take months, and occurs by the formation of fibrous connective tissue to replace the torn ligament, rather than primary healing of the ligament. Conservative management of carpal hyperextension injuries is generally unsuccessful and results in subluxation.⁴⁵ The recommended treatment is panarthrodesis (surgical fusion of all three joint levels) or partial arthrodesis (fusion of only the middle and distal joints), depending on the level of the injuries (Figure 32-13). Rehabilitation is performed as for any arthrodesis, with an emphasis on gait training after the initial healing period (see General Physical Rehabilitation of Joint Arthrodesis). Although the limb function is not normal, prognosis for return to daily activity is good after carpal arthrodesis.⁴⁶ The prognosis is more guarded if instability is caused by rheumatoid arthritis because multiple joints are typically affected.

Carpal laxity syndrome has been described in puppies, which walk with the carpus in a hyperextended position.⁴⁷ The cause is unknown but is speculated to be caused by asynchronous growth between the skeleton and the musculature. Long-term application of supportive splints for hyperextension is avoided because it may lead to muscle atrophy and increased tendon weakness (Figure 32-14). Treatment recommendations generally include balanced diet and controlled exercise on surfaces that provide good

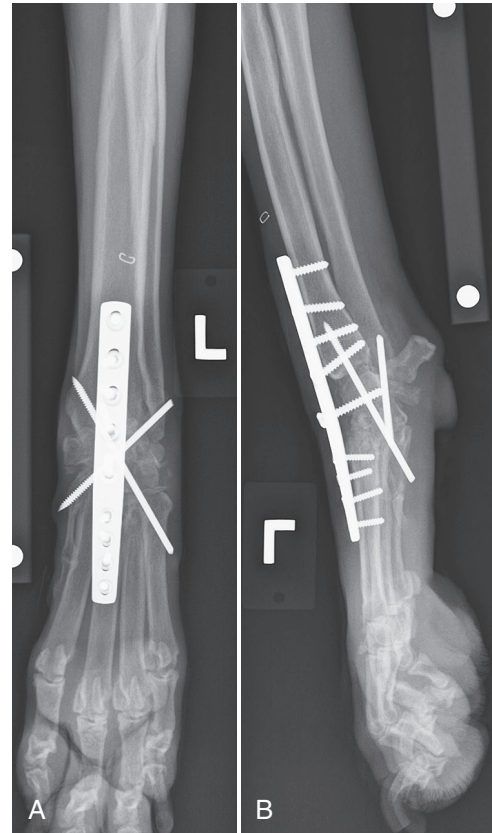


Figure 32-13 Pancarpal arthrodesis performed in a patient with chronic carpal hyperextension. A, A-P view, B, Lateral view.



Figure 32-14 This puppy had increased carpal laxity after long-term use of a splint.

traction. Carpal arthrodesis may be a treatment option for hyperextension that does not resolve with exercise and nutrition, but this cannot be performed until the physes are closed. Carpal laxity syndrome is usually self-limiting, with a short duration and good prognosis.

Carpal flexural contractures have also been recognized in puppies. Poor nutrition has been implicated as a factor, so altering the diet may be beneficial. Splinting may be helpful for puppies that are developing flexural contractures. In severe cases of flexural contracture, tenotomy may be performed.

Trauma of the Superficial and Deep Digital Flexor Tendons

Laceration or avulsion of superficial and deep digital flexor tendons are traumatic injuries that can occur in the front or rear feet. Avulsion or laceration of the tendon insertion on P2 and P3 results in abnormal toe carriage. The tendons may be reattached or repaired for cosmetic reasons and for improved athletic function. Postoperatively the foot is immobilized in a splint for 2 to 3 weeks to limit edema and provide support. Active motion is limited to prevent disruption of the weakened tissue and allow the tendons to heal with minimal adhesions.

Weekly splint changes with ROM exercises of adjacent joints should be performed if possible.³ Passive range of motion and light activity are used to maintain normal tissue length and encourage normal joint movement. Removal of the splint is recommended within 3 weeks of the injury, because prolonged immobilization may delay healing and weaken the repaired tissue. Some tendon motion promotes remodeling of the peritendinous scar, which allows the tendon to glide. However, restricted activity must be enforced, because running or jumping could place catastrophic stress on the healing tissues and result in failure. By 6 weeks the tendons should have achieved adequate strength to sustain normal physiologic stress. At this point the focus of therapy is to restore ROM and strength. If there is contracture of muscle-tendon units, 3-MHz continuous-mode therapeutic ultrasound or hot packs with simultaneous stretching is indicated.³

Luxation of the Metacarpophalangeal and Interphalangeal Joints

Metacarpophalangeal and phalangeal luxations are treated by closed reduction or surgical repair of the collateral ligaments and joint capsule. After repair the foot is splinted for 2 to 3 weeks. It is beneficial to change the splint once weekly so as to perform gentle PROM on the immobilized joints. After the splint has been permanently removed, ROM exercises in the sagittal plane are continued. Limited weight-bearing activities are initiated and gradually increased over another 3 weeks.

Other options for metacarpophalangeal and phalangeal luxations include arthrodesis of the joint or amputation of

the digit. Arthrodesis is a better choice if a high level of function, such as racing, is required. Postoperatively the foot is splinted until there is radiographic evidence of fusion. Weekly splint changes with PROM or mobilization of the other distal joints may be beneficial.

Pelvis and Hip Conditions

Pelvic Fractures

Pelvic fractures account for a large percentage of fractures seen in veterinary practice. Because of the rigid, box-like anatomy of the canine pelvis, trauma usually results in two or more fractures of the pelvis (Figure 32-15, A). In general surgical repair is recommended for fractures within the weight-bearing regions of the pelvis, including the sacroiliac joint, ilial body, and acetabulum. Surgical repair is also recommended if there is neurologic impairment (e.g., possible sciatic nerve entrapment) or medial displacement of fracture fragments impinging on the pelvic canal. Surgical repair results in a quicker return to function with fewer changes secondary to tissue disuse, and the level of function will generally be superior. It is important that surgical repair be undertaken within the first 7 to 10 days after injury to minimize damage to soft tissues and nerves during reduction of the fractures. In some instances, conservative treatment in the form of cage rest may be adequate to allow healing. Examples of injuries amenable to conservative treatment include greenstick fractures of the body of the ilium, mild displacement of the sacroiliac joint with minimal pain or lameness, and fractures of the caudal 20% of the acetabulum. Although pubic and ischial fractures may appear quite severe on a radiograph, these injuries are rarely repaired because they are not part of the weight-bearing axis of the pelvic limb and will heal with rest. Repair of the injuries to the sacroiliac joint, ilium, and acetabulum generally results in adequate reduction and realignment of pubic and ischial fractures for healing.

It is important to identify any concurrent injuries with pelvic fractures. Trauma to the spinal cord or peripheral nerves is relatively common, especially with fractures of the sacrum, fractures of the body of the ilium with medial displacement of the caudal fracture fragment, and fractures of the cranial ischium in the region traversed by the sciatic nerve. It is important to perform a thorough neurologic examination, checking for pain sensation, motor function, and reflexes. Because injuries to the pelvis often involve both pelvic limbs and other injuries to the distal limbs are frequently present, evaluation of the patient while ambulating or performing proprioceptive tests may not be possible. In addition abdominal and thoracic trauma is relatively common in animals with pelvic injuries.

Sacroiliac joint luxations are surgically managed by placing one or more screws in lag fashion to compress the joint. The screw(s) should engage the body of the

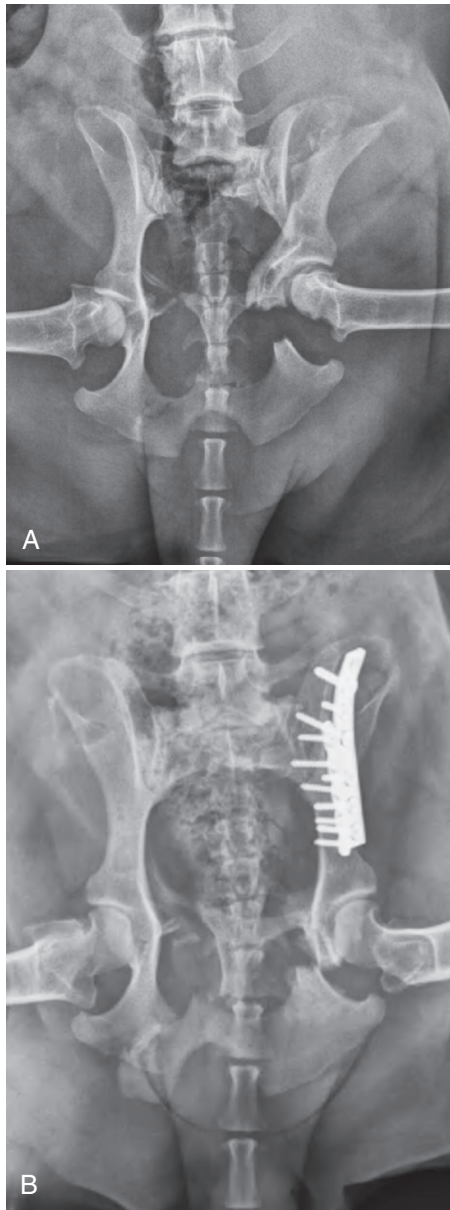


Figure 32-15 The pelvis usually has more than one fracture because of its box-like construction. Typically only the weight-bearing regions of the pelvis are surgically stabilized. **A**, V-D view prior to surgery, **B**, V-D view after surgical stabilization.

sacrum and be deeply seated into the bone, so that the screw engages at least 60% of the width of the sacrum (Figure 32-16). Fractures of the sacral body or wing can be challenging to repair, and concurrent nerve injuries are quite common.

Fractures of the body of the ilium are usually repaired with a bone plate and screws (Figure 32-15, **B**). It is important that the plate be properly contoured to maintain the caudal half of the pelvis in anatomic alignment to avoid future problems associated with a narrowed pelvic canal, such as constipation and obstipation. During the fracture



Figure 32-16 A sacroiliac luxation has also been stabilized with a screw. An ipsilateral hip luxation is repaired using a toggle pin as evidenced by the metal implant medial to the acetabulum and bone tunnel in the femoral head and neck.

reduction, fragments must be carefully manipulated to avoid damage to the lumbosacral plexus, which is located just medial to the body of the ilium.

Fractures of the acetabulum secondary to automobile trauma are common in dogs. Most acetabular fractures should be treated with open reduction and internal fixation. Depending on the size of the patient and the nature of the fracture, repair may involve plates, screws, wires, or polymethylmethacrylate. In many acetabular fractures, anatomic reduction is difficult to achieve, and many animals with these fractures develop OA. Comminuted fractures may be difficult or impossible to reduce, and femoral head and neck ostectomy (FHO) may be necessary. The surgeon should be aggressive in removing as much bone as is necessary, from both the femoral and the acetabular side of the FHO, to prevent bone rubbing on bone postoperatively. It is not uncommon for scar tissue to develop, causing late bony contact in FHOs performed to treat acetabular fractures; therefore early and intense rehabilitation is critical in both preventing and detecting problems before they cause loss of use of the limb (see section on excision arthroplasty).

Rehabilitation should be started immediately postoperatively, even if the animal is nonambulatory. The biomechanical limitations of the fracture repair must be considered, but analgesia, cryotherapy, and passive ROM exercises can usually be initiated immediately postoperatively. On the first postoperative day sling therapy may be

started if the patient is stable. This typically involves short, 5-minute sessions once or twice daily. Advantages of sling therapy include psychological benefits of the patient being in a more normal position, the ability to begin assisted active motion and early, limited weight bearing, and improved access for the caretakers to perform massage and ROM exercises. Rear limb slings can be used with pelvic trauma, whereas whole body slings can be used for patients with forelimb and rear limb trauma. After the incisions are sealed and the patient no longer has intravenous catheters or other invasive lines, water treadmill, or swimming therapy can be considered.

Water treadmill therapy for polytrauma patients is very useful because it allows the therapist to tailor the amount of weight bearing (a water level at the height of the greater trochanter allows the dog to walk with only 38% weight bearing on its limbs). The pressure and buoyancy of the water can also help improve blood flow and decrease edema. Many dogs seem to enjoy being in the underwater treadmill. It can be very demanding for the dog to walk for any length of time after trauma, so the cardiovascular status of the dog should be monitored and very short (several minutes) sessions should be used for the initial treatments.

Hip Dysplasia

Hip dysplasia is an abnormal development of the hip joint, usually bilateral, that occurs primarily in medium and large breed dogs. The causes are multifactorial and include genetic predisposition, rapid growth rate, and diet. Puppies with a genetic predisposition for hip dysplasia develop clinical disease much later in life and to a much lesser extent when fed a restricted calorie diet, as compared with littermates that are fed *ad libitum*.⁴⁸ The hips are normal at birth but develop instability in the early stages of the disease, between 4 and 12 months of age. At this stage the dog may exhibit difficulty rising, decreased activity level, “bunny-hopping” gait, and loss of muscle mass in the hindquarters. Early in the disease process the diagnosis is made by palpation of joint laxity (subluxation). Radiographs may appear normal, although some degree of subluxation may be evident, especially with distraction views (Figure 32-17). As the disease progresses periarticular fibrosis provides some joint stability and the pain may be significantly decreased. However, with further progression of the disease, OA results in pain, crepitus, decreased range of motion, waddling gait, and reluctance to exercise. Thigh and hip muscles atrophy and shoulder muscles may hypertrophy because the body weight is shifted toward the forelimbs. Varying degrees of OA with remodeling of the femoral head and acetabulum are seen radiographically. The rate of progression of hip dysplasia varies between individuals and is difficult to predict. Some dogs may have degenerative changes before 1 year of age, whereas many individuals develop advanced OA in midlife or later.



Figure 32-17 Hip subluxation is typically present bilaterally as a condition of early hip dysplasia. The degree of visible subluxation may be affected by pelvic positioning when making the radiograph.

In young dogs, there are a number of preventive surgical procedures that are designed to change the joint alignment to improve joint stability and slow the progression of OA. The most common of these procedures are the triple pelvic osteotomy (TPO) and juvenile pubic symphysiodesis. Triple pelvic osteotomy is performed on dogs that have early signs of hip dysplasia and joint laxity, but have not progressed to the point of having significant radiographic evidence of OA. Most dogs that fit these criteria are between 4 and 10 months of age. They usually have atrophy of the gluteal and thigh muscles. The technique involves making three osteotomies to mobilize the acetabulum and change its orientation. A bone plate is used to stabilize the ilial osteotomy (Figure 32-18). Postoperatively activity is restricted for 4 to 6 weeks to allow bone healing. Cryotherapy, NSAIDs, PROM, and assisted ambulation, followed by controlled, low-impact therapeutic exercises, are indicated. Leash walking, sit-to-stand exercises, and aquatic walking may be useful to attenuate muscle atrophy while avoiding excessive stress on the repair. After adequate bone healing has occurred the focus of rehabilitation is to strengthen muscles of the hindquarters. Strengthening activities should parallel tissue healing and tissue strength.

Juvenile pubic symphysiodesis is performed in dogs between 16 and 18 weeks of age that are considered to be at risk for developing hip dysplasia. The pubic symphysis is surgically damaged, causing it to fuse and alter pelvic growth. These puppies are often clinically normal at the time of surgery and the surgical trauma is minimal. The focus of rehabilitation is to promote muscular development of the hindlimbs with low-impact exercise.



Figure 32-18 Triple pelvic osteotomy has been performed to improve coverage of the head of the femur by the acetabulum.

Dogs with mild or intermittent signs related to chronic hip OA may be treated by conservative methods to improve function and limit the progression of OA. This includes a combination of NSAIDs, DMOAs, modalities such as heat, electrical stimulation, therapeutic ultrasound, laser, diet and exercise, and is discussed in the section on OA. For dogs having disability or pain that is not adequately managed by conservative methods, there are two salvage surgical options: total hip replacement (THR) or femoral head and neck ostectomy (FHO). Both procedures eliminate the normal joint, thus eliminating the pain. In general, THR is not recommended following an FHO that results in poor function. For rehabilitation after FHO, see the section on excision arthroplasty.

Total hip replacement eliminates the joint, thus eliminating joint-associated pain. The femoral head is removed and the acetabulum is replaced with a prosthetic acetabular cup. A femoral prosthesis is implanted in the medullary canal of the femur. The prostheses may be secured with bone cement, but some systems do not use cement and rely on bony ingrowth for stability ([Figure 32-19](#)). Most dogs have excellent weight bearing almost immediately after surgery. In the initial postoperative period NSAIDs, cryotherapy, and PROM are indicated. One of the most common postoperative complications is hip luxation,⁴⁹ so muscle strengthening after a period of healing is important, especially because there is often preexisting muscle atrophy. Close confinement is enforced when the dog is unsupervised during the first postoperative month. During early ambulation the dog is supported with a sling to prevent adduction or abduction of the limb and dislocation of the prosthesis and inadvertent falls. In some cases, PROM and



Figure 32-19 Cementless total hip replacement performed in a patient with severe osteoarthritis secondary to hip dysplasia.

stretching exercises are indicated to improve hip mobility. Muscle strengthening can be achieved using controlled walking, treadmill activity, and sit-to-stand exercises. The duration of these activities is gradually increased during the first 3 months. Balance and proprioception reeducation may also be important. Dogs are restricted to leash walking, with no running or jumping for the first 3 postoperative months to reduce the chances of implant loosening or dislocation. Prognosis is good to excellent in most cases.

Legg-Calve-Perthes Disease

Legg-Calve-Perthes disease is noninflammatory aseptic necrosis of the femoral head and neck that occurs in small breed dogs. The etiology is unknown, although a genetic component has been identified in some breeds. Dogs usually develop lameness between 6 and 10 months of age. With chronic lameness there is muscle atrophy of the hip and thigh muscles. Radiographs show deformation of the femoral head and neck, with joint incongruity ([Figure 32-20, A](#)). In some cases there may be advanced OA or fractures of the femoral head and neck. If the condition is diagnosed early, a non-weight-bearing sling may be used for 3 to 4 weeks.⁵⁰ However, the treatment of choice is generally FHO ([Figure 32-20, B](#)). Prognosis is good after surgery, provided appropriate rehabilitation is performed beginning immediately after surgery (see [General Physical Rehabilitation Management of Excision Arthroplasty](#)).

Hip Luxation

Coxofemoral luxation is common and usually a result of automobile trauma. However, the presence of hip

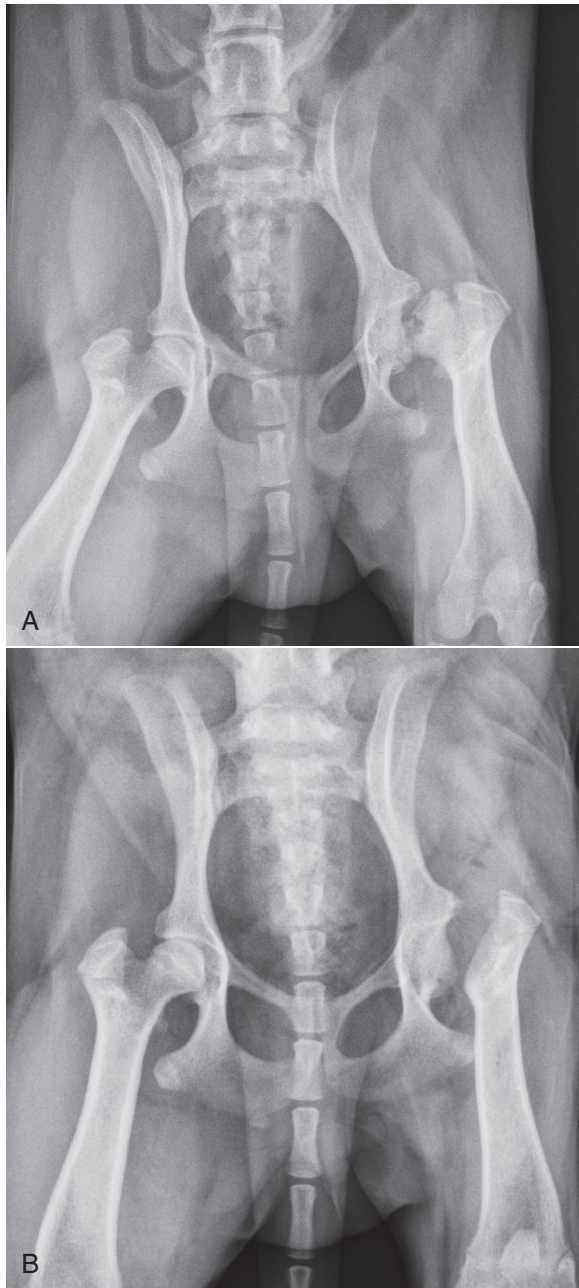


Figure 32-20 A, Young dog with Legg Calve Perthes disease, V-D view. B, Radiograph taken after femoral head and neck excision. Note the complete removal of the femoral head and neck.

dysplasia may diminish joint stability and predispose to luxation with less severe trauma. Hip luxations usually have a craniodorsal displacement. The diagnosis can often be made by palpation of bony displacement, but radiographs are indicated to evaluate for the presence of concurrent fractures or hip dysplasia (Figure 32-21). The presence of these abnormalities may affect the treatment choice and the prognosis.

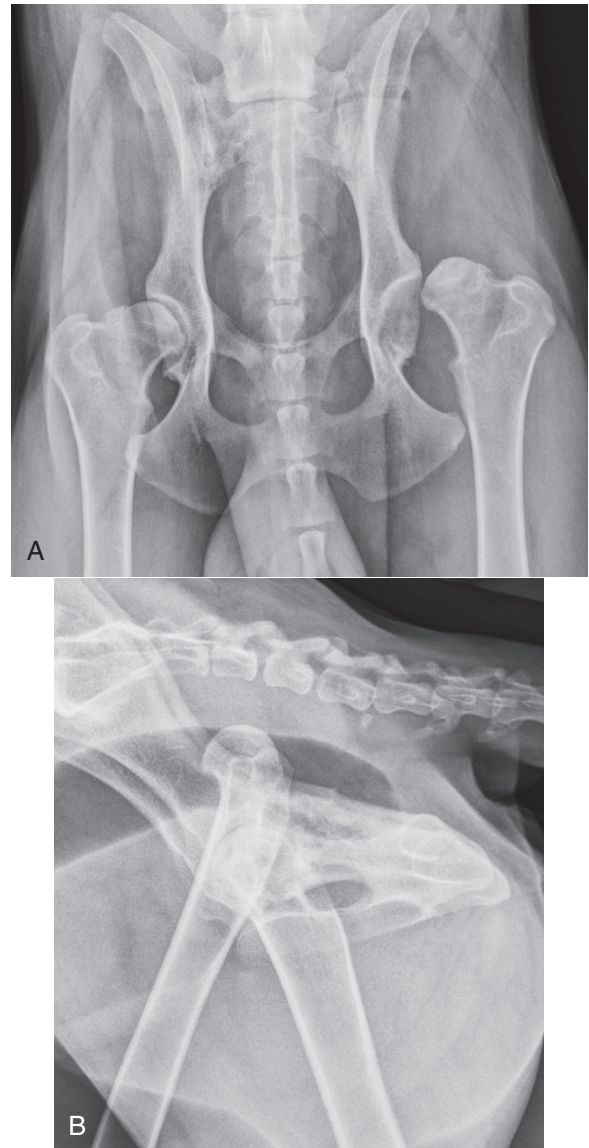


Figure 32-21 Craniodorsal hip luxation in a dog with osteoarthritis associated with hip dysplasia. A, V-D view, B, Lateral view.

If the diagnosis is made within a few days of the injury and there is relatively normal coxofemoral anatomy with no articular fractures, closed reduction and immobilization in an Ehmer sling for 1 to 3 weeks may be successful. An Ehmer sling theoretically maintains the femur in a more stable position of abduction and internal rotation. Eliminating or minimizing weight-bearing activity minimizes stresses on the periarticular tissues to allow them to heal. Open reduction and surgical stabilization are indicated if closed reduction is not successful or articular fractures are present. Common surgical repair techniques include suturing the joint capsule (capsulorrhaphy), creating a prosthetic capsule with suture, or toggle pin fixation (see Figure 32-16). In some cases a combination of techniques may be

used, or the greater trochanter may be relocated more distally and caudally to provide additional stability. After surgery joint healing is often protected by immobilization in the Ehmer sling for 1 to 3 weeks, depending on the degree of tissue damage and the type of repair.

If an Ehmer sling is used, the animal should be evaluated at least weekly to monitor for skin irritation and relaxation. Rehabilitation is begun after the sling is removed. Range of motion must be restored without causing the joint to relaxate. Cryotherapy and NSAIDs are used to minimize inflammation in the early postoperative period. Stretching and PROM activities are also important after hip luxations. Range of motion should be limited to comfortable flexion and extension, with external rotation, abduction and adduction not performed at all during the healing phase of a craniodorsal luxation. Muscle strengthening and re-education begins with walking on a level surface or a downhill grade. Uphill walking, stair climbing, dancing exercises, and aquatic walking may be used to help with hip extension and muscle strengthening during the latter phases of rehabilitation. Walking on an inclined treadmill may promote active hip motion, although the inclination may need to be greater than 5% to cause much increase in hip ROM.⁵¹ In the case of ventral hip luxation, limb abduction should be avoided, and hobbles may be helpful post-reduction to prevent excessive abduction. In all cases activity is gradually increased, but aggressive activities such as jogging, swimming, or uncontrolled play are limited until the joint has healed. This may take 1 to 3 months, depending on the degree of tissue damage.

Joint preservation may not be recommended if there are irreparable fractures involving the joint, hip dysplasia with very poor joint congruency, severe OA, or significant damage to the articular cartilage. In these cases a joint salvage procedure may be a better choice. Common salvage procedures include FHO or THR.

The prognosis for traumatic joint luxations is generally good, although OA is a common sequela. If damage to the articular cartilage is noted at the time of surgery, long-term problems with OA may be expected. If the OA becomes severe, a salvage procedure may be indicated.

Thigh Conditions

Femoral Fractures

Fractures of the shaft of the femur are probably the most common fractures seen in veterinary practice. As such they also have the highest incidence of complications, including delayed or nonunion healing and osteomyelitis. Most complications are related to inadequate fixation or errors in the application of the implants. The possible configurations of femoral fractures vary widely, from a two-piece spiral fracture to a highly comminuted fracture resulting from a gunshot injury. Most of the implants that are available for

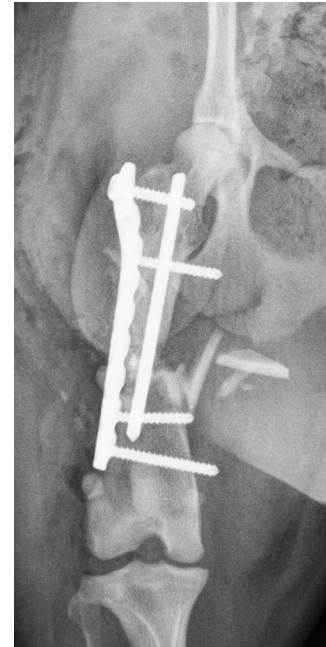


Figure 32-22 Repair of a severely comminuted femoral fracture with a plate-rod technique. A severe fracture configuration suggests a large amount of soft tissue trauma as well.

fracture repair in veterinary medicine may be used to stabilize fractures of the femur, including plates, screws, wires, pins, interlocking nails, and external skeletal fixators (Figure 32-22). Implant failure can result either from single catastrophic overload or from fatigue, implying a greater number of normal weight-bearing cycles than the implant can bear before fracture healing is complete. In general, plates, interlocking nails, and some external fixators tend to be more biomechanically stable than intramedullary pin and cerclage wire constructs.

External coaptation with splints or casts is contraindicated for stabilization of femur fractures because coaptation cannot adequately immobilize the hip joint. In addition coaptation on the limb may only extend as proximally as the mid-thigh, and if so, it can create a fulcrum effect to increase movement at the fracture.

The rehabilitation plan should be based on a number of factors, including the location of the fracture (articular, physeal, long bone), whether more than one limb or bone is involved, pre-existing patient disease (obesity, OA), degree of soft-tissue injury (low energy vs. high energy fracture), the presence or absence of open wounds, and the fixation method(s) used. It is vital that the therapist understand the nature of the fracture and the type of repair that has been applied to know how much activity the dog should have, and what types of complications may occur. For example, if a puppy with a two-piece spiral fracture stabilized with an intramedullary pin and cerclage wires has a sudden loss of limb use with knuckling on the dorsum of the paw and severe pain, the therapist should realize that

the pin may have migrated and entrapped the sciatic nerve. This dog must be treated immediately if there is hope of preventing permanent nerve damage.

Ideally rehabilitation of the fracture patient begins preoperatively, beginning at the time of presentation. Cryotherapy (cold packing) is an essential and simple modality that can be applied in any practice setting. Cryotherapy generally consists of commercial or homemade ice packs that are directly applied over the closed fracture for 10 to 20 minutes, repeated every 2 to 4 hours if practical. Cryotherapy may decrease edema, hemorrhage and inflammation in the soft tissues surrounding the fracture site. Ice packs may be applied during triage of the fracture patient, assuming that body temperature is above 98°F. Temporary stabilization (cast, Robert Jones bandage, or splint) may be applied in some situations if adequate stability can be achieved. The combination of appropriate cryotherapy plus supportive bandaging can dramatically decrease soft-tissue damage and swelling, facilitating surgical repair and decreasing muscle fibrosis postoperatively. As soon as the patient is stable, analgesics should be administered both on a humane basis and to facilitate postoperative rehabilitation by preventing hyperesthesia associated with “wind-up” phenomenon caused by untreated preoperative and intraoperative pain.⁵²

After surgery, the surgeon’s opinion regarding risk of implant failure should be communicated to the rehabilitation team (Figure 32-23). In general NSAIDs, cryotherapy,

PROM, and controlled activities are initiated following surgery. More active forms of rehabilitation such as treadmill walking, UWTM, leash walking, stairs, and swimming are added according to the nature of the injury, method used for fixation, stability of the repair, and temporal aspects of wound and fracture healing. The ROM achieved by the hip joint is affected by the severity of the fracture and the amount of periarticular fibrosis that occurs. Although most dogs develop some degree of OA with fractures of the femoral head or neck, most dogs do reasonably well, although some are candidates for THR or FHO. In addition, if concurrent neurologic injuries are present, appropriate balancing, proprioceptive, and strengthening activities should be initiated.

Femoral Capital Physeal Fractures and Femoral Neck Fractures

Proximal femoral fractures may occur through the capital femoral physis of immature dogs or across the femoral neck of skeletally mature dogs. Rarely do fractures of the articular surface of the femoral head occur. The most serious of the fractures of the proximal femur is the capital femoral physeal fracture, especially in dogs less than 5 months of age. In this fracture the physis may close prematurely, causing a short femoral neck and a small femoral head that results in an incongruent hip joint. In spite of anatomic reduction and internal fixation, these fractures result in OA because of this incongruity.

Femoral capital physeal fractures are typically repaired with either lag screws or divergent K-wires (Figure 32-24). Early repair and minimal soft tissue dissection result in fewer complications. Although they are not generally associated with the muscle contracture that can occur with distal femoral fractures, because of the very small epiphyseal segment and possible tenuous repair, surgeons may occasionally elect to place the limb in an Ehmer or 90-90 sling for 7 to 14 days after surgery. Ideally such slings should be removed as early as possible, and cryotherapy, PROM, and early weight-bearing active ROM should begin within a few days. Collapse of the fracture around the pins can occur. If crepitus, loss of ROM, or increase in pain occur during rehabilitation exercises postoperatively, the surgeon should be contacted immediately or radiographs obtained to assess for loss of reduction or implant penetration into the articular space.

Distal Femoral Physeal Fractures

Although many different repair techniques are described for distal femoral fractures, the most common is cross-pinning. Distal physeal fractures can be difficult to reduce, particularly if they are more than 48 hours old. In addition, distal femoral physeal fractures are predisposed to a devastating postoperative complication characterized by

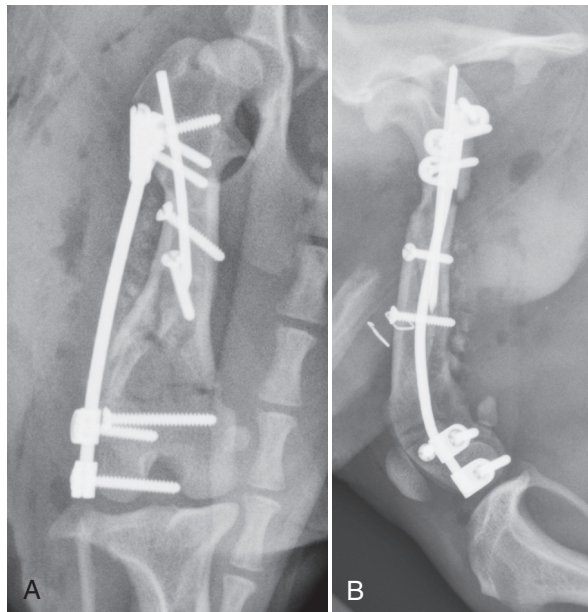


Figure 32-23 Repair of a severely comminuted femoral fracture that involves the stifle joint. **A**, A-P view, **B**, Lateral view. Good communication between the therapist and surgeon are important to understand the estimated stability of the surgical repair. In this case, the presence of a screw in the caudal aspect of the stifle joint may also affect the therapeutic plan.

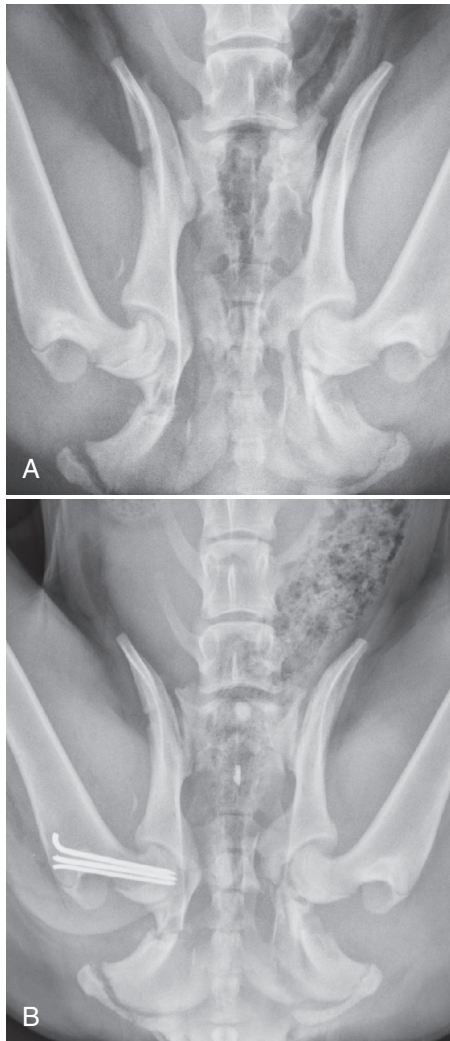


Figure 32-24 Capital femoral physal fracture in a skeletally immature dog. **A**, V-D view prior to surgery, **B**, V-D view after surgery.

quadriceps contracture. A frequent scenario is that the surgical repair is performed, immediately postoperatively the animal seems to have good ROM of the stifle joint, but by the time of suture removal there is excessive stifle extension and severe loss of ROM. Rehabilitation following distal femoral fracture repair is focused on maintaining stifle flexion and hip extension (see [Quadriceps Contracture](#) for more detail on rehabilitation). With a stable repair and good rehabilitation, distal femoral physal fractures are associated with a good prognosis.

Quadriceps Contracture

Quadriceps contracture, also called quadriceps tie-down, is characterized by scarring of the quadriceps muscle group with adhesions to the distal femur, so the stifle is unable to fully flex and the limb is carried in extension ([Figure 32-25](#)). Quadriceps contracture can occur secondary to severe muscle trauma or Neospora, but it is most



Figure 32-25 Inadequate postoperative management a distal femoral fracture led to quadriceps muscle group contracture in this cat.

commonly associated with distal femoral fractures in young animals. The likelihood of quadriceps contracture is greatly increased when the limb is immobilized in extension (e.g., with a cast or splint) following repair of a distal physal fracture. Splinting in extension for as little as 5 to 7 days may result in quadriceps contracture. Chronic disuse can lead to demineralization of the long bones within the affected limb. There may also be disuse atrophy of the articular surfaces of the stifle and periarticular fibrosis. Severe cases can exhibit genu recurvatum or stifle joint ankylosis. Advanced cases of contracture carry a poor prognosis and commonly result in a nonfunctional leg. Surgical release of the fibrous tissue in combination with intense physical rehabilitation can be attempted.⁵³ After fibrosis has occurred attempts to break down the scar tissue by performing ROM exercises, even under anesthesia, may result in refracture of the femur. Surgical release of the contracture without intense rehabilitation will result in reformation of scar tissue within a few weeks. This complication has been successfully managed with a combination of surgical release, dynamic stifle flexion apparatus, and an intensive passive and active ROM exercise program.⁵³ Generally the prognosis is poor and the end result can be an amputation after what initially appeared to be a successful fracture repair.

This problem is best addressed by preventing its occurrence, and prevention of quadriceps contracture depends on several factors. The fracture should be repaired as soon as possible, preferably within 24 hours. Gentle tissue dissection and the use of minimally invasive techniques that minimize tissue trauma (e.g., fluoroscopic guidance of pins) should be employed to facilitate early limb use. Although the ends of the cross-pins may be bent to prevent migration, excess pin protrusion into the soft tissues may cause more pain and inhibit rehabilitation exercises. Some surgeons prefer to countersink the cross pins so there is



Figure 32-26 The 90-90 flexion sling is similar to an Ehmer sling. The goal of the 90-90 is to keep both the stifle and hock joints maintained in 90 degrees of flexion. An Ehmer sling attempts to keep the hip abducted and internally rotated, and the major limb joints are typically fully flexed.

less interference with ROM. The fracture must be anatomically reduced. For any distal femoral fracture, rehabilitation must begin the day of or the day after surgery. If there is any question about the possibility of performing appropriate rehabilitation (e.g., the animal is going home the day after surgery and it is difficult for the owners to manage), the limb should be bandaged in a flexed position to maintain the quadriceps muscle in a stretched position during the acute postoperative inflammatory phase. This can be accomplished by use of a 90-90 flexion sling for 72 hours postoperatively, and this may greatly decrease the risk of contracture (Figure 32-26).⁵⁴ If feasible the sling should be removed on a daily basis and PROM and stretching exercises performed. Cryotherapy can be applied with the sling in place.

If rehabilitative care is readily available, a more physiologic program may be employed. Immediately postoperatively, before anesthetic recovery, cryotherapy is performed and followed by gentle passive ROM exercises. Aggressive treatment of pain, often achieved using a combination of NSAIDs and opioids, is mandatory to successful rehabilitation efforts and this should begin before recovery from anesthesia. After the animal has recovered, cryotherapy is continued three to six times daily. Immediately after cryotherapy, gentle massage is used, working from the distal aspect of the limb toward the proximal limb for 2 to 3 minutes. Although the stifle is the primary area of interest, it is important to treat the whole limb. Therefore gentle flexion and extension of the digits with some stretching is done first, moving up to the hock, stifle, and hip, using 15 to 20 repetitions per joint. For the stifle, it may be helpful to use some gentle stretching (hold the joint at maximal comfortable flexion or extension for 30 seconds). Patella

mobilization and tissue mobilization techniques may be useful to help prevent fibrosis of the various tissue layers to one another. It is important that the process not be painful for the animal and vigorous manipulations are avoided. Cryotherapy is generally continued for at least 72 hours. If the animal resents ROM or tends to hold the limb in extension, a 90-90 flexion sling may be applied intermittently for the first 3 days.

Many animals begin to bear some weight on the limb within 1 to 2 days after surgery, and short, frequent leash walks should be encouraged. Cryotherapy may be replaced with heat therapy after the acute inflammatory phase of wound healing is over, which is typically about 3 to 5 days. Heat causes vasodilation of the cutaneous blood vessels, improves muscle and connective tissue extensibility, and increases the pain threshold. Because of these tissue effects, heat can be very useful before stretching, ROM, and exercise sessions. Some therapists recommend using heat before exercise and cryotherapy after.

Early active exercises should be performed, primarily with very slow leash walks and balancing exercises that encourage active contraction and relaxation of the muscles, and flexion and extension of the stifle. Gentle tug-of-war may encourage limb use, but caution should be used to ensure that excessive force is not placed on the limb. If the animal is leash trained, treadmill walking may cautiously be attempted. It is recommended that an attendant help support the animal and be immediately available in case of resistance on the part of the patient. A syringe cap placed under the contralateral rear foot is also useful to encourage limb use. Finally, swimming is useful to encourage active flexion and extension of the stifle joint, but the incision should be sealed with no drainage or discharge. If the animal is released from the hospital within a few days of surgery, a recheck evaluation should be performed within a few days to ensure that stifle ROM is not becoming reduced. If ROM is preserved and limb function is improving, the animal should be rechecked weekly until the time of radiographic healing, usually by 4 weeks.

Muscle Strain

A strain is a tear or rupture of some part of the muscle-tendon unit, which is caused by excessive stress on the tissues. The injury may occur in the muscle belly, musculotendinous junction, tendon, or teno-osseous junction, but typically affects the musculotendinous region. Mild muscle strain (grade 1) may cause low-grade inflammation with minimal alteration in strength, ROM, or gait. There may be no obvious swelling or palpable pain. Healing may occur within days. With moderate muscle strain (grade 2) there is more damage to muscle and fascia with diminished strength and ROM. Moderate strains generally require one to three weeks to heal. Severe muscle strains (grade 3) are characterized by complete rupture of muscle fibers and fascia, with total loss of muscle function. Healing may take